

Psycho-physiological restoration with audio-visual interactions through virtual reality simulations of soundscape and landscape experiences in urban, waterfront, and green environments

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ABSTRACT

The literature lacks a model of the specific environmental element that induces restoration. Thus, this study contributes to the development of sustainable and healthy environments by examining the effect of the soundscape and landscape experience on psycho-physiological restoration from the audio-visual interaction perspective. It examined various urban and natural environments to propose a restoration model for environmental design application. Sixty participants evaluated three each of urban, waterfront, and green environments using virtual reality technology. The audio-visual characteristics of environments were quantified, and environmental perception was assessed per 14 audio-visual element identifications, six perceived affective qualities, and six overall environmental qualities. The restoration responses were classified as psychological and physiological. Accordingly, 10 restoration models that can interpret design factors of the urban environment with a maximum correlation of 0.80 and an explanatory power of 22 % emerged from urban audio-visual design elements and psycho-physiological responses. Consequently, to improve restoration response, visually attractive and spatial-natural landscapes and acoustically supportive and tranquil soundscapes should be created. Moreover, urban designers can use the models to clearly identify physical characteristics that should be prioritized when promoting the health improvement of urbanites. Furthermore, the study findings can serve as basic data for the development of healthy environments.

1. Introduction

Densely populated urban environments lead to stress and health problems in urbanites (Ulrich et al., 1991; WHO, 2003). Recent research has focused on reducing the negative impact of urban noise—a potent environmental factor that influences human health and well-being—and institutionalizing measures to manage it (Stansfeld, Haines, & Brown, 2000; WHO, 2003). However, understanding urban sound environments from noise and noise-control perspectives does not necessarily enhance life satisfaction. Soundscape refers to sound as a useful cognitive resource (Aletta, Kang, Astolfi, & Fuda, 2016; Hong, Wang, Liu, & Lan, 2019; Kang & Schulte-Fortkamp, 2018). International Organization for Standardization, (2014) defines soundscape as an “acoustic environment as perceived or experienced and/or understood by a person or people, in context.” It provides a new mechanism to interpret the sound

environment from human perception, unlike interpreting the acoustic environment based on physical characteristics. Thus, exposure to nature positively impacts mental and physical health (Berto, 2005; Corazon, Sidenius, Poulsen, Gramkow, & Stigsdotter, 2019; Huang, Yang, Jane, & Bauer, 2020; Kaplan, 1995), and designs of waterfront and green space soundscape elements are used to create sustainable and healthy cities (Collins et al., 2020; Grahn & Stigsdotter, 2010; Jeon, Lee, You, & Kang, 2010; Jeon, Lee, You, & Kang, 2012; Wolch, Byrne, & Newell, 2014).

To understand restoration responses through environmental experiences, we consider two representative theories: attention restoration theory (ART) (Kaplan & Kaplan, 1989) and stress recovery theory (SRT) (Ulrich et al., 1991). These theories share the assumption that natural environments provide psycho-physiological restoration for urbanites. However, ART assumes that exposure to natural environments restores cognitive resources (e.g., attention and memory) and

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psycho-physiological energy, while SRT assumes that exposure to natural environments induces a positive emotional experience through which psycho-physiological energies are recovered. First, regarding observational studies, Van den Berg, Maas, Verheij, and Groenewegen (2010) concluded that access to a green environment serves as a buffer against stress and induces mental recovery. Thompson et al. (2012) found that providing a natural environment within urban spaces positively impacts mental health by evaluating the amount of cortisol secretion. Fan, Das, and Chen (2011) discovered that vegetation in urban spaces promotes stress reduction and social support. Mitchell (2013) concluded that regular experience of the natural environment and activity in that space significantly reduces mental health deterioration. Gerber et al. (2019) noted that virtual reality (VR) natural environment experience based on the ART theory could have restoration effects; they found that it affects the parasympathetic nervous system and reduce physiological-response-related stress. Indeed, observational studies commonly found that a positive environment increases human activity and effectively reduces stress. Next, in aspects of experimental design studies, the restorative effect of positive soundscapes in soundscape ecology has been actively studied from the perspective of ART (Aletta, Oberman, & Kang, 2018; Jeon, Jo, & Lee, 2021) and may be reviewed separately based on psychological and physiological effects. Psychological effect surveys have shown that positive soundscapes effectively reduce stress, reinforce positive feelings (Krzywicka & Byrka, 2017; Shu & Ma, 2020), and improve cognitive function (Korpela & Ratcliffe, 2021). Natural soundscapes have positively changed physiological responses as measured through diverse physiological indicators (e.g., heart rate, heart rate variability, skin conductance level, and electroencephalography; Hume & Ahtamad, 2013; Li & Kang, 2019; Medvedev, Shepherd, & Hautus, 2015; Yu, Lee, & Luo, 2018; Deng et al., 2020; Erfanian, Mitchell, Aletta, & Kang, 2021; Li, Ba, & Kang, 2021). Li and Kang (2019) found that natural soundscapes decreased heart rate, respiratory frequency, and respiratory depth, and increased R-wave, heart rate variability, and α and β wave amplitudes. The few studies that have investigated soundscapes' restorative effects have shown inconsistent findings (Aletta et al., 2018). Therefore, our study aimed to fill this gap and analyze the potential effects of soundscape and landscape on physiological indicators.

It is crucial to consider the complex effects of multisensory soundscape ecology (Hong & Jeon, 2015; Truax & Barret, 2011), including the impact of audio-visual interactions on soundscape perception (Carles, Barrio, & de Lucio, 1999; Carles, Bernáldez, & de Lucio, 1992; Jeon & Jo, 2020; Li & Lau, 2020; Preis, Kociński, Hafke-Dys, & Wrzosek, 2015). Studies have examined the impact of soundscape and landscape perception in independent investigations (Jeon & Jo, 2020; Liu, Kang, Behm, & Luo, 2014). Moreover, the impact of audio-visual interactions on environmental quality has been investigated using descriptors, such as comfort, satisfaction, pleasantness, and tranquility (Jo & Jeon, 2021; Liu & Kang, 2018; Pheasant, Fisher, Watts, Whitaker, & Horoshenkov, 2010). The impact of audio-visual interactions on environmental quality can be further examined as follows. Carles et al. (1992) found that preference for urban environments is influenced more by combinations of sounds rather than visuals, and providing a sound environment appropriate for the landscape that comprises each space can enhance the emotional well-being of urbanites. Moreover, Carles et al. (1999) found that the congruency of landscape and soundscape is a factor that significantly affects space preference, while Galbrun and Calarco (2014) also concluded that providing a natural view that goes well with the acoustic environment is a way to enhance environmental preference. A follow-up study (Jo & Jeon, 2020) reconfirmed that the appropriateness of audio information and visual information is an important factor that affects space preference. Liu, Kang, Luo, and Behm (2013) discovered that the identification of sound source is more influenced by the actual landscape view of the space than the space's intrinsic characteristics. Preis et al. (2015) found that manipulating the audio element is more effective than the visual element when deciding upon the audio-visual

comfort of the urban environment. Jeon and Jo (2020) found that perception of the space differs per the presence or absence of audio and visual information, and when both information is provided simultaneously, the space's naturalness is perceived in a novel way. Jo and Jeon (2021) found that providing green-related elements visually increases the pleasantness of sound, while providing water-related elements visually can control the eventfulness of sound. Lu, Hasegawa, Tan, and Lau (2022) observed that visual distraction influences the pleasantness of crowd sound and suggested that the perception of soundscape may differ per the combination of visual views and sounds. A comprehensive review related to audio-visual interaction by Li and Lau (2020) shows that the perception of the urban environment is influenced by audio or visuals and clear interactions. This study employed a wide-ranged review and observed that the audio component has a greater effect than the visual component on environmental preference. In some cases, this notion was utilized in design based on audio-visual interaction. It is possible to enhance noise satisfaction by providing a natural element to the visual scene. Further, the annoyance urbanites psychologically feel can also change per the composition of the visual material envelope (Hong & Jeon, 2014). Tatlikazan, Poetter, Hannemann, Bürkle, and Schulte-Fortkamp (2017) suggested that expectations about the soundscape of a space can change given the visibility of building structures, social structures, and sound sources. The influence of such audio-visual interaction is not limited to external spaces. The negative influence of various noise sources can be reduced even for indoor spaces, such as offices, with the provision of a water-scape. Thus, it is essential to consider the interaction between visual and auditory elements when studying environmental space perceptions.

Recently, disciplines other than soundscape ecology have investigated the impact of environmental audio-visual interactions on psycho-physiological restoration (Deng et al., 2020; Hedblom et al., 2019; Jiang et al., 2021; Li et al., 2021). To summarize, extant research has examined the differences in the restorative effect of visual and auditory information, and according to a space's function (e.g., nature and city). However, a model explaining which specific environmental elements (e.g., audio-visual and perceived affective quality) induce restoration is lacking. Moreover, the lack of ecological validity is a clear limitation, with most studies performing evaluations on a 2D screen in a laboratory environment (Orland, Budthimedhee, & Uusitalo, 2001; Pitt & Naasauer, 1992). Contrastingly, we used VR technology to simulate real-world conditions, thereby reducing the gap between theory and practice. Human cognition and restoration responses were investigated to develop a practical perception model that can be applied when designing cities promoting psycho-physiological restoration (Aletta & Xiao, 2018).

The current study aimed to comprehensively investigate the effect of soundscape and landscape experiences on psycho-physiological restoration responses from the viewpoint of audio-visual interactions and to propose a restoration model that could be directly applied to space design. As noted, extensive research has explored the positive impacts of nature, soundscapes, and landscapes to design comfortable environments for urbanites. Even so, this topic remains confined to academic research that is challenging to use as basis material for urban designers and stakeholders. This situation stems from the limited number of variables in most studies and the limitation in proposing a model that can conduct general interpretation given the narrow research scope. Further, researchers have employed different design factors for varying purposes, making it challenging to generalize and implement research results. Researchers have also provided statistically complex models, making them challenging for practical use. This study has a more general scope and introduces an integrated model that offers intuitive and accessible insights for urban designers, planners, and decision-makers based on various variables (physical characteristics, psychological response, physiological response). That is, it proposes a model that can be practically used to prioritize and make decisions on factors that enhances urbanites' mental and physiological health.

2. Methods

2.1. Site selection and audio-visual data collection

Nine sites in Korea, representative of both urban and natural environments (waterfront and green space), were selected for evaluation. The panoramic view and detailed audio-visual characteristics of each site are shown in Fig. 1. For urban environments, three sites in downtown Seoul functioning as high-density commercial (Gangnam-daero), low-density commercial (Insa-dong), and business areas (Jongno-gu) were chosen. For waterfronts, sites with a river (Han River), wetland (Suncheon Wetland), and ocean (Yeosu Odong Island) were selected. Finally, for green space, a forest (Boseong Green Tea Fields), a valley (Jogyesan Provincial Park in Suncheon), and a temple (Seonamsa Temple in Suncheon) were selected. Consequently, nine evaluation points have been selected for this study. Regarding landscape, they include 1) urban areas with abundant artificial elements, 2) natural areas with significant waterfront features, and 3) natural areas rich in various landscaping elements. Regarding soundscape, they include 1) urban areas with considerable artificial element-related sounds (e.g., traffic noises and music), 2) natural areas dominated by water sounds, and 3) natural areas with considerable birdsongs. While waterfront and green spaces are in the same natural environment category, waterfront spaces are defined by topographical features, and green spaces are areas with sounds from biodiversity, such as various vegetation, animals, and insects.

The details of the methodology are described in Appendix A and Appendix B (Fig. B.1 and Table B.1) to allow for a replicable laboratory experiment set-up with VR techniques and physical characteristics of the audio-visual stimuli.

2.2. Experimental design

2.2.1. Participants

Soundscape perception differs per individual characteristics, such as health conditions, personality traits, preferences, and experiences (Jeon et al., 2021; Lercher, van Kamp, von Lindern, & Botteldooren, 2016; Lindborg & Friberg, 2016; Sun et al., 2018; Wilkie & Stavridou, 2013). Psychologically stable individuals evaluated urban soundscapes

positively on pleasantness. Such differences in perception induced differences in restoration (Jeon et al., 2021). Wilkie and Stavridou (2013) identified environmental preference as an important factor determining restorative effect, and Korpela and Ylén (2009) concluded that restorative effect is greater in individuals in poor health. von Lindern, Bauer, Frick, Hunziker, and Hartig (2013) demonstrated that the level of a space's restorative effect varies per the purpose of the visit; the restorative effect was greater in those visiting a place to reduce stress. Jeon et al. (2021) demonstrated that character (acquired through experiences), rather than temperament (which is inherited), is more critical in assessing restorative potential.

To minimize variations in perception and psycho-physiological responses in the sample, only students living in Seoul and attending a university or graduate school located in the metropolitan city were recruited into the study. It was reasoned that with students, it would be possible to obtain a relatively controlled study sample, as students' life patterns (i.e., going to class and working in a laboratory) are relatively homogenous compared with the general population. Appendix C (Table C.1) summarizes participants' individual characteristics. The participants' ages ranged from 21 to 30 years (mean age, 24.3; standard deviation, 2.4), and all had typical vision and hearing. Participants who participated in this study had an equal gender ratio of 30 males and 30 females. Participant recruitment announcements were posted on message boards and social network services of the universities located in Seoul. As this study aimed to investigate the impact of experiencing various environments on psycho-physiological restoration, one inclusion criterion was physically healthy people with mild depression, stress, or anxiety. Those experiencing higher levels of depression, stress, or anxiety requiring medical treatment were excluded from the study.

The adequacy of the sample size was estimated using G-power software (version 3.1.9.4). The minimum sample size required was estimated at $n = 42$ by assuming a moderate effect size of 0.5, a significance level of 0.05, and a power of 0.8. Of the 70 students who participated, 10 were excluded on the grounds of their health status on the day of testing (e.g., lack of sleep, fatigue, and health problems) or outliers in the data (e.g., flight response to a stress stimulus or a score of three or more standard deviations away from the mean). The excluded participants did not participate in the experiment. Finally, data from 60 participants were analyzed, exceeding the minimum required sample size.

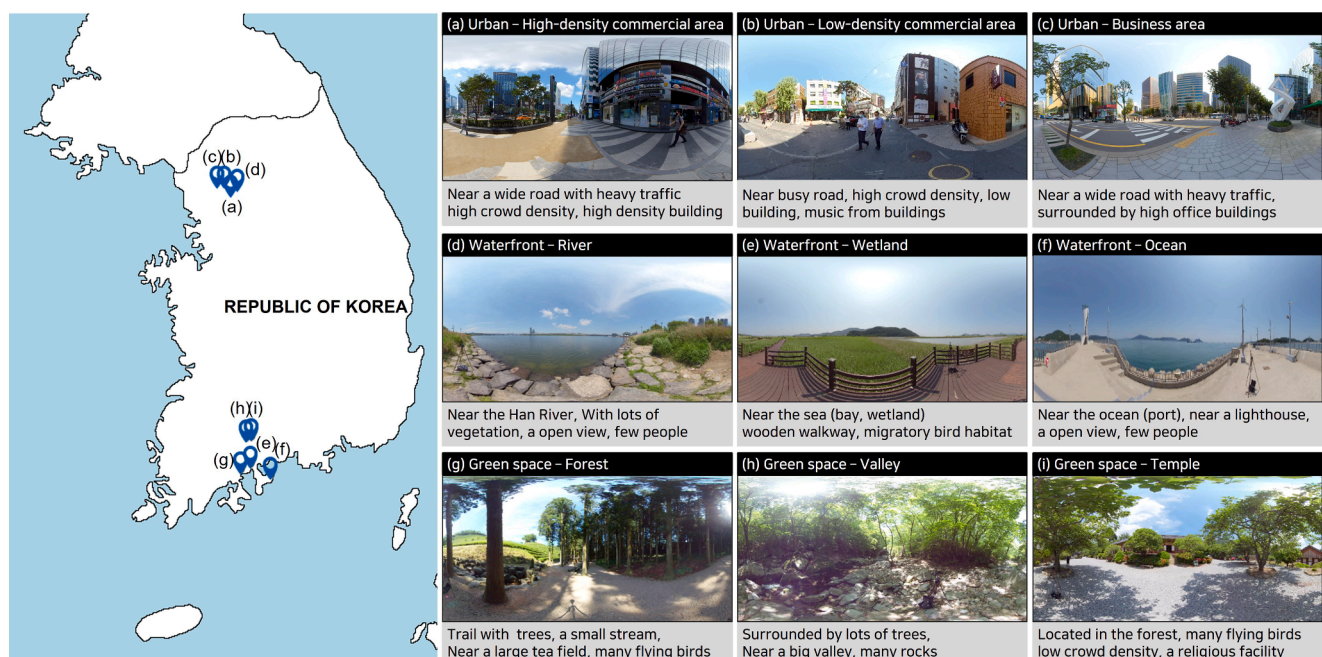


Fig. 1. Panoramic views and details of the nine evaluation sites: (a) – (c) urban, (d) – (f) waterfront, (g) – (i) and green areas.

2.2.2. Procedure

The evaluation procedure and sessions are illustrated in Fig. 2 (a). The study was approved by the Institutional Review Board of Hanyang University (No. HYUIRB-202,010–011). The final sample of 60 participants committed to a two-day experiment. All experiments were conducted in the same laboratory, and the internal temperature was maintained at 23 degrees Celsius. Psycho-physiological restoration responses to various environments were examined on Day 1, and environmental perceptions concerning soundscapes and landscapes were examined on Day 2. The participants were advised to sleep well and rest fully on the previous day and they were instructed to abstain from smoking, drinking, and consuming caffeine or drugs, which could alter their physiological responses during the experiment. Before commencement, the participants were provided with a brief written document describing the study's purposes and the theories on restoration through soundscape and landscape, and they were fully informed of the use, treatment, and storage of the scale and physiological response data. Written consent was then obtained from the participants. A questionnaire assessing demographic characteristics was created using Google Forms and administered to the participants. The participants completed their questionnaires before the experiment began. Detailed information on the procedure is available in Appendix D.

2.2.3. Stress task

Although the participants recruited for the study were all students aged 20–29 years, individual participants could likely present with different emotional states before exposure to the audio-visual environments. Accordingly, before the presentation of the audio-visual stimuli, controlling their emotional state was necessary. A stress task was administered before they were exposed to the audio-visual environments to artificially induce a state of stress, as the study aimed to investigate the capacities of psycho-physiological restoration responses. In consideration of the study design of repeating experiments using several audio-visual stimuli, the mental arithmetic stress test (MAT) was chosen to induce stress (Giles, Mahoney, Brunyé, Taylor, & Kanarek, 2014). The MAT induced stress by showing the participants a mental subtraction problem for a second in the VR environment. An example question included “repeatedly subtract 23 from 1000 and state the answer aloud as you mentally perform each subtraction.” The participants were given three seconds per question and provided with correct/incorrect feedback each time. To confirm whether stress is induced during a stress task, the activities of the hypothalamus-pituitary-adrenal (HPA) axis and the sympathetic-adrenal-medullary system were assessed either directly (using bodily fluids) or indirectly (via heart rate variability or HRV). In this study, HRV was measured to confirm stress induction (Ulrich-Lai & Herman, 2009). The low-frequency (LF) to high-frequency (HF) ratio, one of the HRV indicators, was used to verify that the MAT task used in this study induced stress in participants. The LF/HF ratio shows the activity levels of the sympathetic and parasympathetic nervous systems, and the normal range is between 0.5–2. When the HPA axis is stimulated, cortisol levels increase, increasing the LF/HF ratio. The average LF/HF ratio of the participants before testing was normal at 1.7, but participants handling stress tasks showed an average LF/HF ratio of 2.7, which surpasses the normal range. Hence, the MAT task selected for this study induced stress in the participants.

2.2.4. Psychological responses

To collect subjective responses (including psychological responses) to the audio-visual environment, a questionnaire was constructed with the items shown in Table 1 (Appendix E). There were three categories (individual characteristics, restoration, and environmental perception) in the questionnaire. Detailed items in each of the categories were selected by referring to previous studies (Bradley & Lang, 1994; Cohen, Kamarck, & Mermelstein, 1983; Herranz-Pascual et al., 2019; Jeon & Jo, 2020; Lee et al., 2012; Liu et al., 2013; Min, Lee, Kim, Suh, & Kim, 2000; Min, Oh, & Lee, 2007; Payne, 2013; Wang, Long, Chen, & Li,

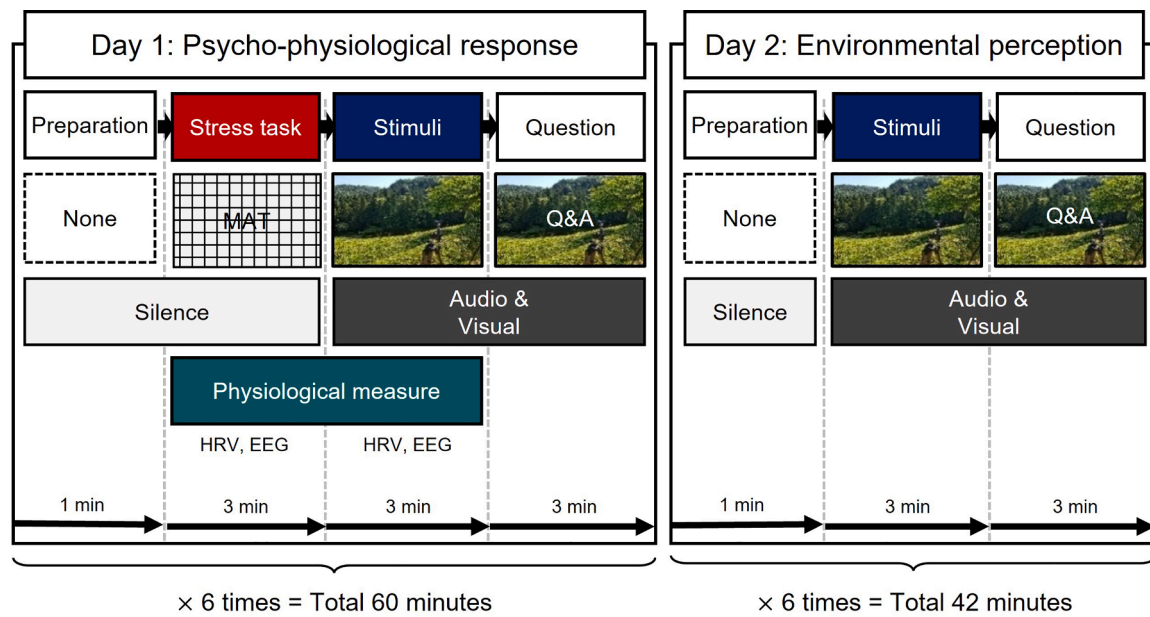
2019; Watson, Clark, & Tellegen, 1988; Weinstein, 1978; Zhao, Xu, & Ye, 2018).

2.2.5. Physiological responses

To investigate the physiological restoration response according to the audio-visual environment experienced, HRV and electroencephalogram (EEG) were selected as indicators (see Fig. 2 [b]). HRV is known for objectively and reliably assessing activity in the autonomic nervous system. The autonomic nervous system's activity changes in response to changes occurring inside and outside of the body, and HRV measures cyclical variations in the heartbeat across time. HRV is linked to the interplay between the sympathetic and parasympathetic nervous systems on the sinus node. In this study, it was measured for three minutes using SA-3000NEW (Medicore, Korea), with sensors attached to the inside of both wrists and the inside of the left ankle. For HRV measurement, five parameters were chosen and defined as follows: 1) total power (TP; unit: ms^2) indicates the overall activation level of the autonomic nervous system, 2) the standard deviation of normal to normal (SDNN; unit: ms) refers to the standard deviation of R-R intervals in the sinus rhythm and indicates the level of stress resistance, 3) temporary stationarity index (TSI) indicates the level of pressure to the human body based on parameters like heart rate (HR) and HRV and is a stress index, 4) LF (unit: ms^2) refers to HRV on a low-frequency spectrum (0.04–0.15 Hz) and is used as an index of sympathetic activity and is suggestive of the level of fatigue, and 5) HR (unit: bpm) is the mean number of heart beats per minute. The physiological indicators used in this study are defined as follows. First, TP reflects the regulatory ability of the autonomic nervous system. That is, when there is stress, TP decreases relative to the state during good health. SDNN is the HR variability, which shows the body's ability to maintain a stable state when adapting to changing environmental conditions. TSI represents the HR and variability influenced by external activities. The higher the TSI value, the greater the fatigue. LF is related to mental stress and decreases with fatigue, showing how much body energy is lost.

Brain waves may be defined through cortical or scalp recordings of electrical brain activities generated as ions transmitted through the biochemical interactions between brain cells. Brain waves were measured with 32-channel EPOC Flex (EMOTIV, USA). Electrode conductive paste, Ten20 (Opaque, high-viscosity, salty), was used to obtain smooth signals, and electrodes were attached following the 10–20 System of Electrode Placement, an internationally standardized protocol. Like HRV, brain waves were measured for three minutes. A quantitative analysis was performed on brain waves using EmotivPRO software (version 2.6.2.302) by setting a sampling rate of 128 Hz and a bandwidth of 0.2–45 Hz (high attenuation at 50 Hz and 60 Hz). Before the main analysis, the data were preprocessed to remove ocular artifacts in EEG signals. Artifacts were removed with visual inspection and manual independent component analysis to identify and separate independent sources from superimposed signals using Fast Fourier Transformation with a four-second window and a two-second overlap. EEG signal processing and source-level feature extractions were conducted in $\alpha 1$ (8.00–9.75 Hz) using iSyncBrain® by iMediSync.¹ It has been reported that α waves (α -EEG, 8–12 Hz) commonly occur in a comfortable state of self-acceptance or during meditation. Further, low- β waves (low- β -EEG, 12–16 Hz) indicate a midway state between relaxation and tension and emerge while the person is aware of their surrounding environment, explores, and performs a new task, whereas high- β waves (high- β -EEG, 16–25 Hz) emerge while the person is in an aroused state, focused with eyes open, or in a state of anxiety and excitement. In this study, the α/β ratio (ABR), computed by dividing α -EEG by the mean of low and high β -EEGs, was used as a brain wave index (Luijckx, Vossen, Hermens, van Os, & Lousberg, 2015; Riera et al., 2012). The ABR shows us the relaxation response to the brain's arousal

¹ Retrieved October 1, 2019 from <https://isyncbrain.com>



(a) Procedure

**ECG Function Specification**

- Lead: 3 leads
- Heart Rate measurement range: 30 ~ 200 BPM
- Accuracy: ± 2 BPM
- Frequency response 1: Digital band-stop filter on (notch filter)
- Frequency response 2: Digital band-stop filter on (50/60Hz)
- Input Signal Range: $\pm 0.05 \sim 5$ mV peak

**EEG Function Specification**

- Number of channels: 32 (plus CMS/DRL reference)
- Sampling method: Sequential sampling (Single ADC)
- Sampling rate: 128 SPS (1024 Hz internal)
- EEG Resolution: 14 bits 1 LSB = $0.51 \mu\text{V}$
- Bandwidth: 0.2 – 45 Hz (high attenuation at 50 Hz and 60 Hz)
- Filtering: Built in digital 5th order Sinc filter
- Dynamic range: ± 4.12 mV

(b) Equipment for physiological response measurement

Fig. 2. (a) Procedure of the experiment over two days: Psycho-physiological restoration responses were measured on Day 1, and environmental perceptions were assessed on Day 2; (b) equipment image and specification for physiological response measurement.

response. The higher the ABR value, the more relaxed the state of an individual.

2.3. Data analysis

2.3.1. Normalization of physiological response

Owing to individual differences in physiological responses, there is a limitation in comparing absolute values; hence, all physiological responses were standardized (Li & Kang, 2019). The relative change (%) between the physiological responses observed for three minutes during the stress task (baseline) and the exposure to a stimulus (raw data) was computed for standardization, whereby a standardized physiological response value was calculated using the following formula: $((\text{raw value} - \text{baseline value}) / \text{baseline value}) \times 100$. With this standardization, it was easy to analyze the extent of physiological restoration from the stress

state when the participants were exposed to audio-visual stimuli. Standardization enabled comparison across stimuli as well as across participants, as it is based on relative change.

2.3.2. Statistical analysis

The data were analyzed using SPSS Statistics (IBM Corp., Version 25.0) as follows. First, normality (Shapiro-Wilk and Kolmogorov-Smirnov test) and homoscedasticity (Levene) assumptions were tested. The normality assumption was met, and parametric tests were performed accordingly. Analysis of variance (ANOVA) was performed to test the significance of the differences in the psycho-physiological restoration effect among the three types of space. Moreover, the Pearson correlation between each indicator was analyzed to examine the relevancy between psychological and physiological responses. A canonical correlation analysis (CCA) was conducted to examine the

Table 1

Three-category questionnaire: restoration and environmental perception.

Category	Number	Scale	Reference
I. Individual characteristics			
Gender	2	M / F	-
Health state			
PSS for stress condition	10	0 – 4	(Cohen et al., 1983; Lee et al., 2012)
Sensitivity for noise	21	1 – 6	(Weinstein, 1978)
Demographic factors			
TCI-RS	140	1 – 5	(Min et al., 2007)
WHOQOL-BREF	26	1 – 5	(Min et al., 2000)
URPS	20	1 – 5	(Wang et al., 2019)
II. Restoration			
Psychological restoration			
PRSS for restorative effect	19	1 – 7	(Payne, 2013)
SAM for mood state	3	1 – 9	(Bradley & Lang, 1994)
PANAS for mood state	20	1 – 5	(Watson et al., 1988)
III. Environmental perception			
Soundscape perception			
Sound source identification	7	1 – 5	(ISO 12913-2, 2018; Jeon & Jo, 2020)
Perceived affective quality	14	1 – 5	(Herranz-Pascual et al., 2019; ISO 12913-2, 2018)
Overall soundscape quality	3	1 – 5	(ISO 12913-2, 2018; Jeon & Jo, 2020)
Landscape perception			
Visual element identification	7	1 – 5	(Zhao et al., 2018)
Perceived affective quality	14	1 – 5	(Jeon & Jo, 2020)
Overall landscape quality	3	1 – 5	(Liu et al., 2013)

Note: PRSS: Perceived Restorativeness Soundscape Scale; SAM: Self-Assessment Manikin; PANAS: Positive Affect and Negative Affect Scale; PSS: Perceived Stress Scale; WHOQOL-BREF: World Health Organization Quality of Life – Brief Version; TCI-RS: Temperament and Character Inventory – Revised Short Version; URPS: Urban Residents Psychological Security.

relationships of psycho-physiological responses with physical characteristics and environmental perception. CCA is an extension of multiple regression and correlation analyses; it explains the relationship between two sets of multidimensional variables by converting them to canonical variate pairs in a lower dimensional space (Hotelling, 1935; Thompson, 1984). If multicollinearity is likely to be present because of a great number of input variables, it is preferable to interpret the relationship between two sets of variables using canonical loadings rather than canonical correlation coefficients. Hence, in this study, canonical loadings were used to interpret canonical analysis results. Although canonical loadings are stable in the presence of multicollinearity, they are used only to compare the relative degree of correlations between canonical variables and original variables in a set as they do not reveal the correlation structure between the former in one set and the latter in the other. To overcome this problem, canonical cross-loadings were also used in interpreting the CCA results.

3. Results

3.1. Environmental perception

Regarding the perception of audio-visual elements of the soundscapes and landscapes, the mean responses on sound source identification are visually illustrated in Fig. 3 (a). In the graphs, dominance percentage refers to the proportion of participants who rated 3 or higher on a five-point Likert scale for the element under evaluation (Jeon & Jo, 2020). In the urban environment, artificial sound elements, such as traffic noise, stood out with a percentage of over 50 %. In the natural

environment, the perception of natural sound elements (birdsong, water, and wind sounds) was high in both the waterfront and green space, being 76–97 %. Mean responses regarding visual element identification are presented in Fig. 3 (b). In the urban environment, the perception of artificial visual elements like vehicles and buildings stood out, 45–47 %, while the perception of people was 23–26 %, showing a trend similar to sound source identification. In the natural environment, the perception of natural visual elements (vegetation, water, and sky) was high at 56–73 %. The perception of open space was relatively high in the natural environment, 20–28 %, unlike the corresponding perception in the urban environment, which was 11–12 %.

The results on the perceived affective quality of the soundscape are presented in Fig. 3 (c), indicating mean factor scores by the function of the spaces. To test for statistical significance in the difference in the mean factor scores according to function, one-way ANOVA and Turkey posthoc tests were performed; the results are visually illustrated in Fig. 3 (c). Given various noises in the urban environment, the mean factor score for *tranquil soundscape* was -0.94 , lower than that of the natural environment, while the mean factor score for *detrimental soundscape* was very high, being 0.82 . The differences were statistically significant ($p < 0.01$).

Regarding the natural environments, the mean factor score of a waterfront for *tranquil soundscape*, 0.19 , was slightly higher than that of a green space. Green space had a high factor score for *supportive soundscape*, 0.22 , and a low factor score for *detrimental soundscape*, 0.16 . However, these differences were not statistically significant. From the soundscape perspective, it can be argued that waterfronts are calm while green spaces are pleasant. Urban and natural environments did not show statistically significant differences for *supportive soundscape*. The participants responded more sensitively to negative than positive elements in evaluating the soundscape in urban and natural environments. The results for the perceived affective quality of the landscape are shown in Fig. 3 (d). Compared with natural environments, the factor score for *attractive landscape* was extremely low (-0.59) in urban environments, while that for *discomforting landscape* was high (0.57). The differences were statistically significant. The factor score for *spatial/natural landscape* was extremely low (-0.81) in urban environments due to high-rise buildings and vehicles downtown, whereas the corresponding score for waterfront with panoramic views (such as ocean and wetland) was extremely high (0.87). Although green space could have been perceived as somewhat enclosed because of the surrounding vegetation, the factor score for *spatial/natural landscape* was at a level midway between the corresponding scores for urban and waterfront landscapes (-0.03), probably due to a high proportion of natural elements.

Mean responses regarding the overall quality of the soundscapes and landscapes are visualized separately according to the function in Fig. 3. To test for statistical significance in the differences in overall environmental quality according to function, one-way ANOVA and Turkey posthoc tests were performed. Regarding the overall quality of the soundscapes, all mean values of overall impression, appropriateness, and sound satisfaction were statistically significantly lower in the urban environment than in the natural environment, as shown in Fig. 3 (e). Regarding the overall quality of landscapes, most mean values of scenic beauty and visual satisfaction in the natural environment were 4.1 or higher, being statistically significantly higher compared with the urban environment (2.7 – 2.9 ; Fig. 3 (f)). The mean value of the status of infrastructure was 4.1 in the urban environment, being statistically significantly higher than in the natural environment (2.5 – 3.5). Among the natural environments, the mean value of the status of infrastructure was slightly higher for waterfront than for green space, by about 0.3 . This is speculated to be because, as is evident from Fig. 1, the infrastructure in the downtown area is relatively better in the natural environment (such as the Han River) than in the rural environment.

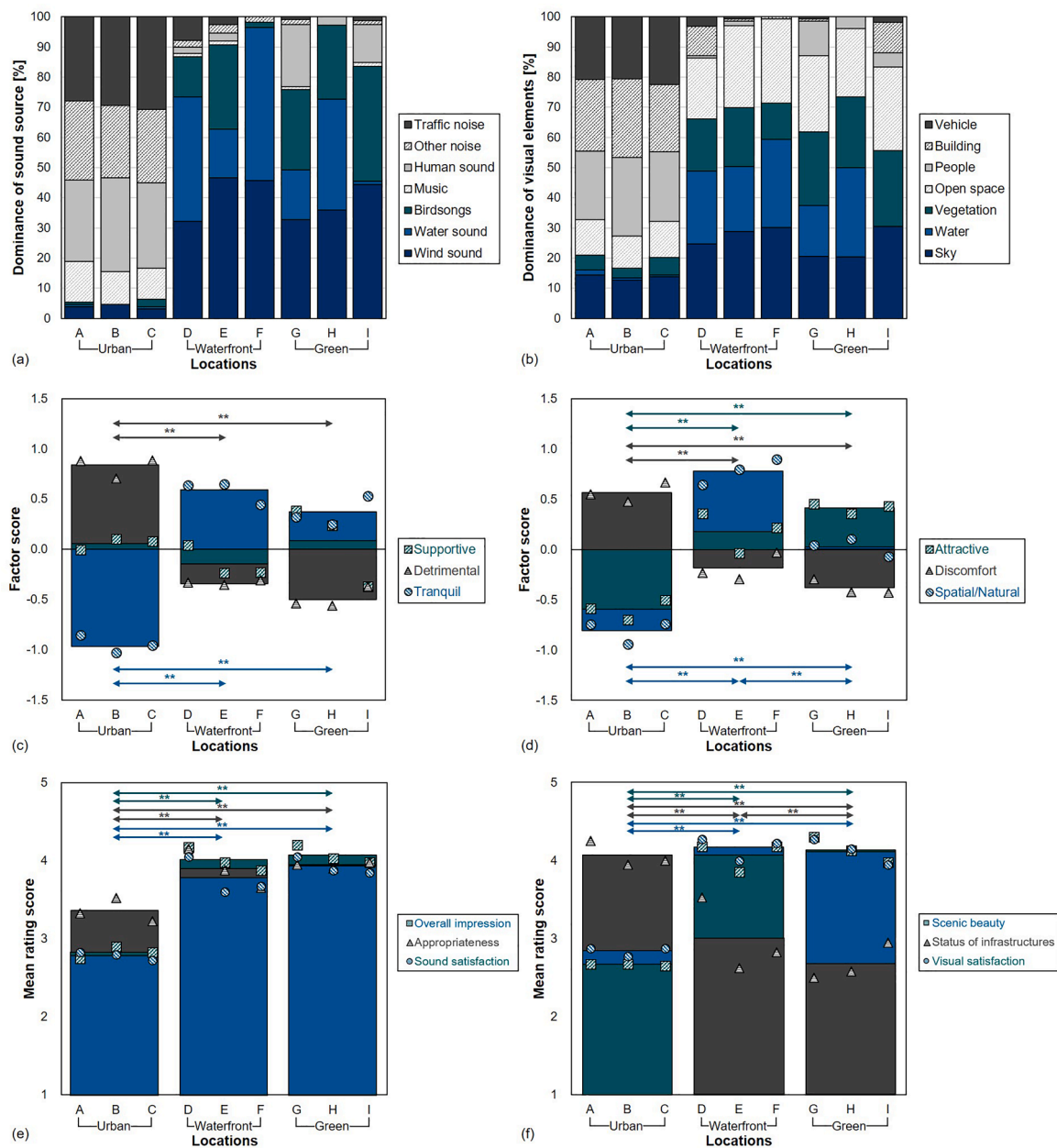


Fig. 3. Summary of subjective responses for environmental perceptions. The dominance of (a) sound source and (b) visual element; perceived affective quality of (c) soundscape and (d) landscape; overall quality of (e) sound environment and (f) overall quality of the visual environment. **p-value < 0.01.

3.2. Psycho-physiological responses

3.2.1. Psychological restoration

The psychological and physiological aspects of restoration through the experience of soundscapes and landscapes in diverse environments were examined separately. Fig. 4 shows the psychological restoration responses obtained using various scales. To test for the differences among reference states (the absence of stimuli; refer to section 2.3.5.), urban, waterfront, and green space, one-way ANOVA and Tukey posthoc analyses were performed. Regarding attention restoration, the mean Perceived Restorativeness Soundscape Scale (PRSS) scores of the six domains are visually illustrated in Fig. 4 (a). The mean PRSS score was 4.41 in the natural environment and 2.92 in the urban environment, indicating a greater psychological restoration effect in the former. As fascination is a form of involuntary attention, we believe that in the natural environment, the participants were able to focus easily without depleting their mental resources.

“Breaking away” refers to the feeling of escaping from a fatigued cognitive structure and finding rest. This occurs when another thought structure is activated via a physical or conceptual transition brought on by a particular experience. Generally, “breaking away” can be categorized as “escaping to” and “breaking away from.” “Escaping to” describes moving to a new place or adopting a new thought that distracts from an unpleasant or tiring experience. “Breaking away” refers to the feeling of withdrawing from a specific place or thought and finding relief. Compared to the reference, “breaking away” was very low in the urban environments at 1.93, and extremely high in the natural environments, with 4.91 in the waterfront, and 5.02 in the green space. Within all the domains, “breaking away” showed the largest differences, which were statistically significant. Accordingly, it was reasoned that in the urban environments, participants felt somewhat confined in their everyday lives, whereas in the natural environments, they felt like they were escaping from or being freed from their daily lives and responsibilities.

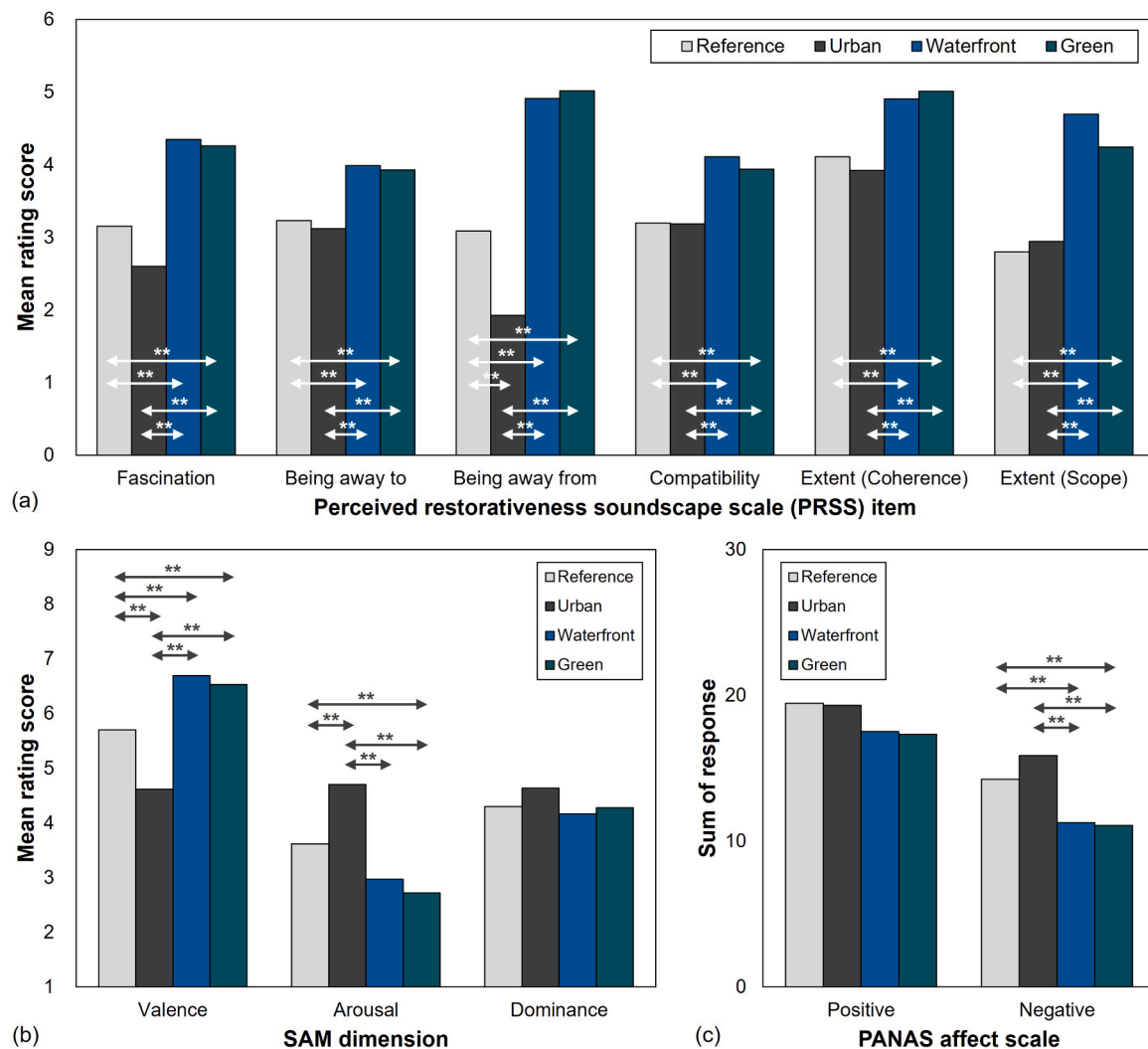


Fig. 4. Mean rating score of psychological responses: (a) PRSS (six domains), (b) Self-Assessment Manikin (SAM) (three domains), and (c) Positive Affect and Negative Affect Scale (PANAS) (two domains). ** represents p -value < 0.01 for analysis of variance analysis between different environments. The difference in psychological response for the Dominance and Positive according to environmental change was not statistically significant.

Compatibility refers to the level of harmony between nature and individuals. As compatibility increases, there are fewer discrepancies to resolve in the environment, and, hence, individuals may not need to use focused attention, eventually experiencing psychological restoration. Consequently, compatibility was higher in the natural environment than in the urban environment, being 0.93 and 0.76 for waterfront and green space, respectively. Thus, it is speculated that involuntary attention consumption was lower in the natural environment, as in fascination. Extent refers to consistency and richness in the environment, classified into coherence and scope. In coherence, consistency in an environment is evaluated based on the connection among environmental elements, and in scope, an environment is evaluated on the dimension of size. Again, the scores were higher in the natural environment than in the urban environment, and, in particular, the extent (or scope) showed a difference of at least 1.30. This finding is consistent with those reported above in that the responses regarding spatial/natural landscapes were much higher for the waterfront landscapes than in the urban environments.

The results of the Self-Assessment Manikin (SAM) and Positive Affect and Negative Affect Scale (PANAS), which examined emotional change, are visually illustrated in Fig. 6 (b) and (c), respectively. The SAM results were analyzed separately for each of the three domains (valence, arousal, and dominance). Valence refers to pleasure. Compared to the

mean valence score of the reference (5.7), the corresponding score was statistically significantly lower in the urban environment (4.6) and higher in the natural environment (6.7 in the waterfront and 6.5 in the green space). However, the mean score for arousal, a state of activation and excitement, was the highest in the urban environment (4.7) and was statistically significantly higher than the waterfront (3.0) and green space (2.7). Compared to the reference, the mean valence score was statistically significantly reduced in the green space, suggesting that participants became calmer. Dominance (the SAM domain referring to the level of control over or freedom from emotions) showed little difference according to function, even when compared to the reference. All dominance scores were around 5, the midpoint in the SAM response scale. Therefore, dominance did not seem to be affected by the function.

The PANAS results were examined separately for positive and negative affect, as shown in Fig. 4 (c). Compared to the reference, the urban environment did not differ for either positive or negative affect, indicating that participants did not show noticeable emotional responses. In the natural environment, however, the negative effect was statistically significantly lower than in the reference. This suggests that participants' cognitive-behavioral tendencies expanded in natural environments, and they were able to maintain a state of psychological stability. Interestingly, positive affect varied little according to function, somewhat inconsistent with the valence responses in the SAM. This

could be attributed to the inability of the verbal descriptors in the PANAS to fully describe the feelings experienced at each site. The participants may be more sensitive to negative elements, as demonstrated earlier in the perceived affective quality showing a significant difference in the *detrimental soundscape* but not in the *supportive soundscape*.

3.2.2. Physiological restoration

Physiological data on restoration response during exposure to soundscapes and landscapes are shown in Fig. 5. HRV was analyzed using the five indices, TP (autonomic nerve activities), SDNN (stress resistance), TSI (stress index), LF (fatigue), and mean HR. The results are visualized for each spatial function in Fig. 5 (a). Regarding TP, of the natural environments, exposure to waterfront showed a large rate of increase, over 10.5 %, compared to the reference, and the increase was statistically significant. Such a positive effect of waterfront in physiological change was also confirmed in stress-related indices, wherein SDNN increased by 13.9 % and TSI decreased by 9.4 %. Similar findings were observed in the green space, where SDNN increased (14.6 %) and TSI decreased (9.5 %) compared to the reference. This confirmed that in the natural environment, the participants were well-adjusted to the environmental change and were able to maintain a stable state. Stress reduction physically and mentally affects the level of fatigue, and overall, LF was decreased in the natural environment, although not significantly. Interestingly, SDNN significantly increased in urban environments compared to the reference. This may be because the participants were students living in an urban area and were therefore able to adjust to the environmental change as they were familiar with the urban stimuli. HR increased statistically significantly in all environments compared to the reference. Generally, HR tends to accelerate in the presence of a stimulus in quiet surroundings (Irwin, Hall, Peters, & Plack, 2011) and has been shown to increase both in more pleasant and eventful environments (Erfanian, Mitchell, Kang, & Aletta, 2019; Hume & Ahtamad, 2013; Medvedev et al., 2015). Hence, it is speculated that in this study, HR increased in all environments because of various reasons.

Lastly, the α/β ratios (ABR) based on brain wave measurements were analyzed by the space's function, as shown in Fig. 5 (b). Compared to the reference, ABR increased in both urban and natural environments, regardless of function. The analysis of mean values in each frequency spectrum showed a small difference of below 4 % in the increase rate in the α wave between urban and natural environments, whereas β wave (the mean of low and high β s) was higher in natural than urban environments by 5–8 %. This is consistent with Li and Kang's (2019) finding

that nature sounds increased α and β waves. Accordingly, in this study, ABR was lower in the natural environment than in the urban environment. However, because the differences in brain wave data were not statistically significant, it may be inferred that the participants were in a stable psychological state irrespective of the type of space. This study examined only the overall pattern in brain waves; future research must explore the details regarding activity and connectivity in each brain region.

Further, to check whether the changes in the physiological response confirmed in this study are significant relative to the changes in the physiological response because of daily activities, it was compared with previous studies. Yi et al. (2021) showed that HRV parameters, such as HF, TP, and HR, change within the range of 1 % to 6 % when performing meditation or simple gymnastics for the elderly. Meanwhile, the EEG response was shown to change by 12 % for α wave and 12 % for β wave. Moreover, when forest walking activity was performed, the HRV response changed by 4 % to 17 %, and the α and β waves changed by 13 % and 12 %, respectively. Takahashi et al. (2005) showed that, when playing meditating roles, LF/HR decreased by 16 %, and α and β waves increased by approximately 6 and 4 %, respectively. That is, the change in HRV and EEG responses through the soundscape experience in this study were similar or significantly changed relative to other daily activities. Therefore, the restoration of the soundscape is a crucial factor to address in the urban design process.

The Pearson correlation between each indicator was analyzed to examine the relevancy between psych-physiological responses, shown in Table 2. First, the psychological response shows that PRSS indicators have a statistically significant positive correlation with the positive domains of PANAS and the valence domain of SAM. However, the negative domain of PANAS and the arousal domain of SAM show a statistically significant negative correlation. Physiological response shows that TP, SDNN, and ABR have similar relevancies, while TSI, LF, and HR show similar patterns. That is, if stressed, the activity of the sympathetic nervous system decreases, which decreases stress resistance and reduces psychological stability. Meanwhile, the stress index or fatigue increases while the HR decreases. The relevancy between psychological response and physiological response shows that the “being away to” and “being away from” indicators of the PRSS indicators show a weak but statistically significant positive relationship with TP and SDNN while showing a negative relationship with TSI. That is, the sense of liberation the environment can give humans can also induce psychological recovery and the vitalization of the sympathetic nervous

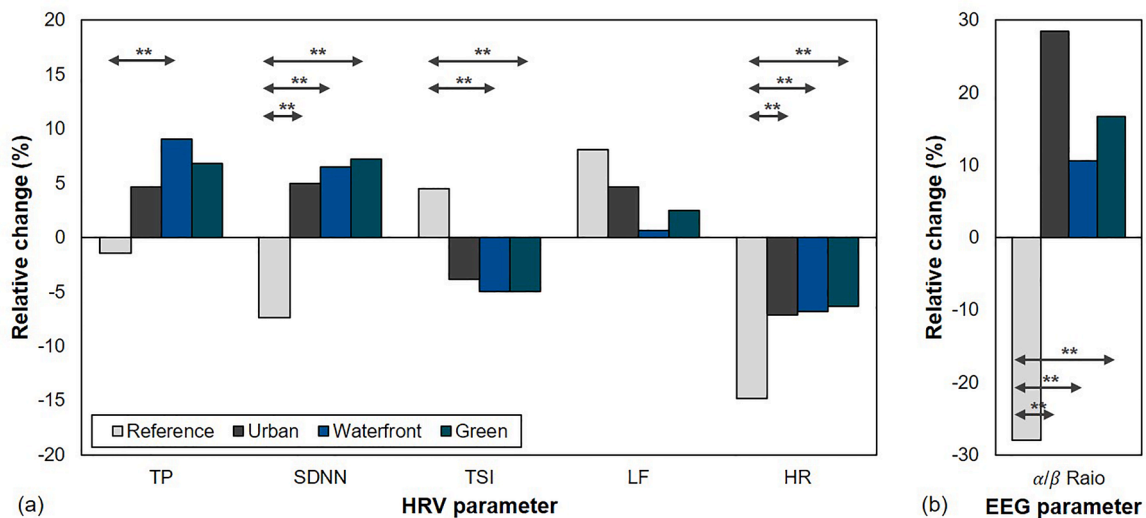


Fig. 5. Summary of physiological response: (a) heart rate variability (HRV) and (b) electroencephalogram (EEG). ** represents p-value < 0.01. The difference in physiological response for the low frequency (LF) per environmental change was not statistically significant. TP: total power; SDNN: standard deviation of normal to normal; TSI: temporary stationarity index; HR: heart rate.

Table 2
Pearson correlation analysis among psycho-physiological response.

Variables	Fa	BAT	BAF	Com	ExC	ExS	Pos	Neg	Val	Aro	Dom	TP	SDNN	TSI	LF	HR	ABR
Psychological response																	
PRSS																	
Fa																	
BAT	0.53**																
BAF	0.74**	0.53**															
Com	0.76**	0.69**	0.62**														
ExC	0.64**	0.49**	0.62**	0.61**													
ExS	0.38**	0.51**	0.59**	0.61**	0.49**												
Pos	0.26**	0.35**	0.46**	0.37**	0.10*	0.09											
Neg	-0.38**	-0.18**	-0.45**	-0.23**	-0.24**	0.31**	0.31**										
Val	0.71**	0.46**	0.66**	0.57**	0.54**	0.32**	0.21**	-0.54**									
Aro	-0.28**	-0.18**	-0.36**	-0.17**	-0.26**	-0.17**	0.26**	-0.43**	-0.43**								
Dom	0.07	0.11*	0.07	0.09	0.06	0.04	0.07	0.07	0.07	0.07							
TP	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07						
SDNN	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08					
TSI	-0.05	-0.16**	-0.08	-0.06	-0.10*	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01				
LF	-0.02	-0.04	-0.07	-0.04	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01	-0.01			
HR	-0.01	0.03	-0.01	0.03	-0.01	0.03	-0.01	0.03	-0.01	0.03	-0.01	0.03	-0.01	0.03	-0.01		
ABR	-0.08	-0.01	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03	-0.03
Physiological response																	
HRV																	
TP	0.07	0.16**	0.14**	0.08	0.09	0.03	-0.04	-0.02	0.04	-0.01	-0.04	0.73**	0.73**	-0.70**	-0.66**	-0.10*	0.11*
SDNN	0.08	0.21**	0.11*	0.07	0.09	0.01	0.11*	-0.07	0.06	-0.04	-0.01	0.73**	0.73**	-0.82**	-0.45**	0.05	0.14**
TSI	-0.05	-0.16**	-0.08	-0.06	-0.10*	-0.01	0.06	0.01	-0.03	-0.01	-0.02	-0.70**	-0.82**	0.47**	0.47**	0.18**	-0.11*
LF	-0.02	-0.04	-0.07	-0.04	-0.01	0.02	0.03	-0.03	0.02	0.01	0.04	-0.66**	-0.45**	0.17**	0.17**	0.17**	-0.03
HR	-0.01	0.08	0.04	-0.01	0.03	0.04	-0.12*	-0.07	-0.01	-0.03	0.04	-0.10*	0.05	0.18**	0.17**	0.01	0.01
EEG																	
ABR	-0.08	0.01	-0.07	-0.04	-0.03	0.01	-0.08	0.06	-0.07	0.05	0.04	0.11*	0.14**	-0.11*	-0.03	0.01	0.01

Note: Fa: Fascination; BAT: Being away to; BAF: Being away from; Com: Compatibility; ExC: Extent Coherence; ExS: Extent Scope; Pos: Positive; Neg: Negative; Val: Valence; Aro: Arousal; Dom: Dominance; PRSS: Perceived Restorativeness Scale; SAM: Self-Assessment Manikin; PANAS: Positive Affect and Negative Affect Scale; HRV: Heart Rate Variability; EEG: electroencephalogram; TP: Total Power; SDNN: Standard Deviation of Normal to Normal; TSI: Temporary Stationarity Index; LF: Low Frequency; HR: heart rate; ABR: α/β ratio. * and ** represent $p < 0.05$ and $p < 0.01$, respectively.

system, which leads to physical recovery effects.

3.3. Restoration model based on audio-visual interactions in various environments

To propose a restoration model from the perspective of audio-visual interaction, CCA was performed to investigate the relationships of psycho-physiological responses to physical characteristics and environmental perception. Through CCA analysis, the study derived the primary model with the greatest correlation between psycho-physiological response, physical characteristics, and environmental perception, after which an additional canonical variate model revealed a strong correlation while being independent of model 1. This process was repeated, and only models evaluated to be statistically significant were presented in Fig. 6 and Table 3 (p -value < 0.05). For each model, the corresponding proportions of variation are summarized in Table 3. Canonical loadings are shown in Fig. 6. For a clear understanding of the CCA results, the findings were interpreted by focusing on the variables with a canonical loading of 0.5 or higher.

3.3.1. Relationship between physical characteristics and psycho-physiological restoration

As shown in Fig. 6 (a), there was a strong correlation (0.80) between the physical characteristics and psychological restoration factors. Among the physical characteristics factors, L_{Aeq} (sound strength), L_{A5-A95} (temporal variation), loudness, water ratio, and grey ratio were identified as important, while among the psychological restoration factors, “escaping to,” valence, fascination, arousal, and negative affect were similarly identified. As L_{Aeq} , L_{A5-A95} , and loudness decreased in an auditory environment and as the grey ratio decreased and water ratio increased in a visual environment, psychological facets of fascination, “breaking away,” and valence increased, and arousal and negative affect were reduced. Regarding explanatory power, 29 % of the variance in the psychological restoration was explained by the set of canonical variables for the factor, and 19 % by physical characteristics. The correlation between physical characteristics and physiological restoration was not significant and, thus, not presented ($p = 0.982$).

3.3.2. Influence of audio-visual identification on psycho-physiological restoration

The relationship between the audio-visual element identification factor and the psycho-physiological restoration factor is shown in Fig. 6 (b)–(c). The correlation between the factors was 0.78, as can be seen in Fig. 6 (b). Of the facets of psychological restoration, “breaking away,” valence, fascination, and “escaping to” increased. Of the emotional facets, arousal, and negative affect decreased. Meanwhile, the audio-visual elements like artifact-related (traffic noise, music, vehicle, and building) and human-related (human sound and people) elements decreased, and nature-related elements (wind sound, water space, water sound, and birdsong) increased. Regarding explanatory power, 29 % of the variance was explained by the set of canonical variables for psychological restoration and 17 % by audio-visual identification. The correlation between the audio-visual element identification factor and physiological restoration factor was not high (0.33), as shown in Fig. 6 (c). The eigenvalue was very low (0.13), and the canonical loadings of audio-visual elements were also low—all under 0.5. The explanatory power of the set of canonical variables for physiological restoration was 16 %, and the explanatory power of audio-visual element identification was 2 %.

3.3.3. Effects of perceived affective quality on psycho-physiological restoration

The relationships between the perceived affective quality factors and psycho-physiological restoration factors are shown in Fig. 6 (d)–(g). A CCA model structure of the three statistically significant models was extracted from the factors. Regarding the first canonical variate pair in

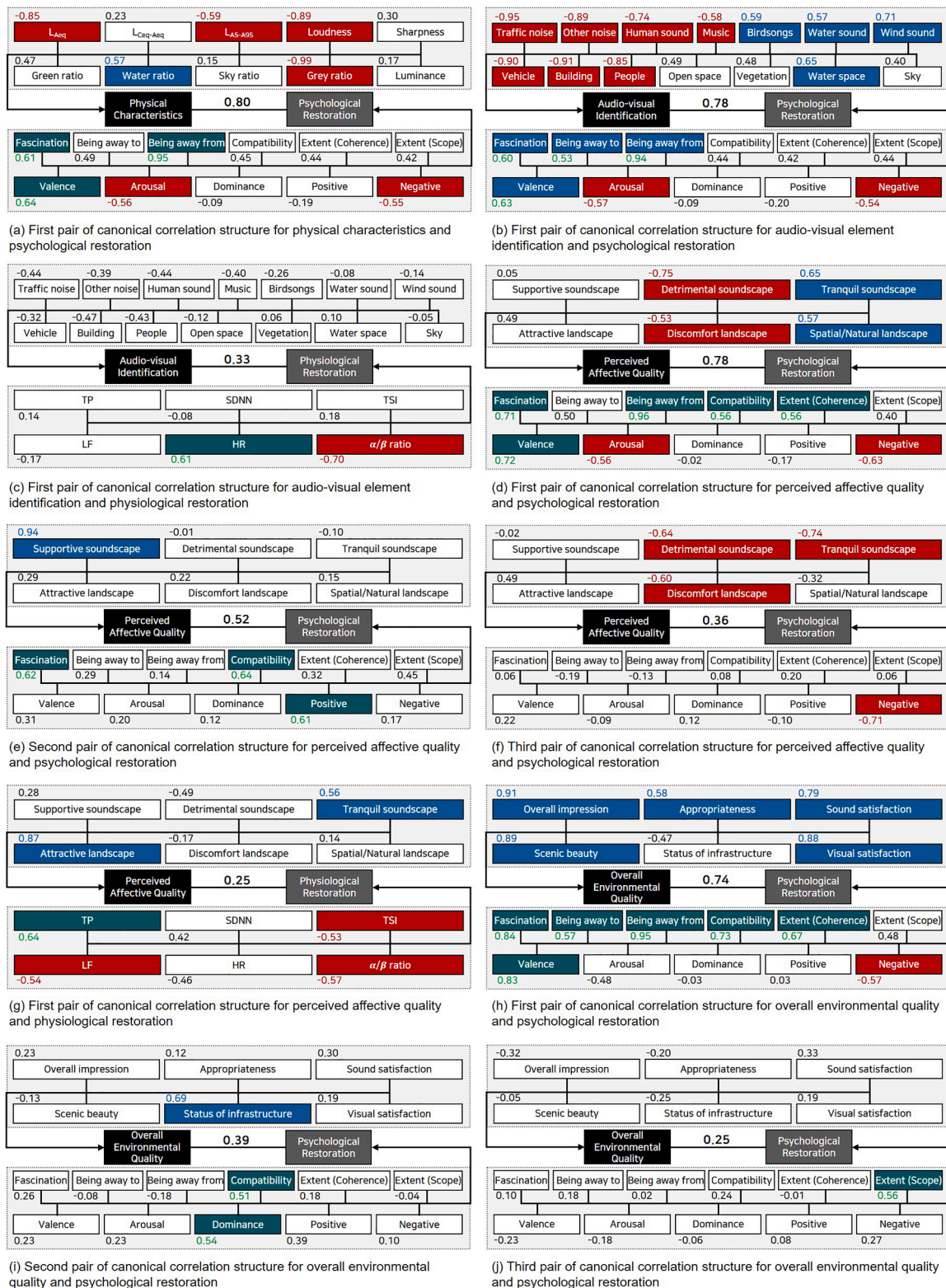


Fig. 6. Restoration model with canonical loadings through canonical correlation analysis. (a) – (j) refers to models (a) – (j).

the canonical correlation structure, the correlation between the perceived affective quality factor and psychological restoration factor was high (0.78), as shown in Fig. 6 (d). The positive psychological restoration response increased as the *detrimental soundscape* and *discomforting landscape* decreased, and the *tranquil soundscape* and *spatial/natural landscape* increased. In the second pair, the *supportive soundscape*

was identified as an important variable in perceived affective quality, as shown in Fig. 6 (e), and compatibility, fascination, and positive affect increased as the *supportive soundscape* increased. In the third pair, *detrimental* and *tranquil soundscapes*, and *discomforting landscapes* were identified as important variables in perceived affective quality, as shown in Fig. 6 (f). Results for the relationship between perceived affective

Table 3

The proportion of variance obtained for different sets of structures.

Model	Eigenvalue	Set 1 by self	Set 1 by set 2	Set 2 by self	Set 2 by set 1	Sig.
Physical characteristics ↔ Psychological restoration						
1	1.83	0.36	0.23	0.29	0.19	p < 0.01
Physical characteristics ↔ Physiological restoration						
1	-	-	-	-	-	p = 0.98
Audio-visual identification ↔ Psychological restoration						
1	1.53	0.51	0.31	0.29	0.17	p < 0.01
Audio-visual identification ↔ Physiological restoration						
1	0.13	0.09	0.01	0.16	0.02	p < 0.05
Perceived affective quality ↔ Psychological restoration						
1	1.58	0.31	0.19	0.34	0.21	p < 0.01
2	0.37	0.18	0.05	0.16	0.04	p < 0.01
3	0.15	0.25	0.03	0.06	0.01	p < 0.01
Perceived affective quality ↔ Physiological restoration						
1	0.07	0.24	0.02	0.28	0.02	p < 0.05
Overall environmental quality ↔ Psychological restoration						
1	1.22	0.60	0.33	0.40	0.22	p < 0.01
2	0.18	0.11	0.02	0.09	0.01	p < 0.01
3	0.07	0.06	0.01	0.05	0.01	p < 0.05
Overall environmental quality ↔ Physiological restoration						
1	0.08	-	-	-	-	p = 0.19
Individual characteristics ↔ Psycho-physiological restoration						
1	1.05	0.04	0.02	0.11	0.05	p < 0.01
2	0.48	0.05	0.02	0.06	0.02	p < 0.01
3	0.39	0.04	0.01	0.07	0.02	p < 0.01
4	0.28	0.10	0.02	0.03	0.01	p < 0.01
5	0.24	0.06	0.01	0.05	0.01	p < 0.01
6	0.20	0.05	0.01	0.06	0.01	p < 0.05

Note: Sig.: Significance.

quality and physiological restoration are presented in Fig. 6 (g). These factors showed a relatively weak relationship of 0.25. *Tranquil soundscape* and *attractive landscape* were identified as important variables to the perceived affective quality factor; in the physiological restoration factor, TP, TSI, LF, and ABR were identified as important. Increases in the dimensions of perceived affective quality had positive physiological restoration effects. The explanatory power of perceived affective quality was low, with 2 % of the variance of psychological restoration explained by the factor.

3.3.4. Influence of overall environmental quality on psycho-physiological restoration

The relationship between the overall environmental quality factor and the psycho-physiological restoration factor is presented in Fig. 6 (h)–(j). A CCA structure of three models was extracted. In the first canonical variate pair, the overall environmental quality factor and psycho-physiological restoration factor showed a high correlation of 0.74, as shown in Fig. 6 (h). In overall environmental quality, all variables were identified as important, except for the status of the infrastructure, and potentially facilitative of psychological restoration. The explanatory power of overall environmental quality was very high, with 22 % of the variance of psychological restoration explained by the factor. The

correlations in the second and third pairs were low, 0.39 and 0.25, respectively. It was observed in the second pair that compatibility and dominance increased as the status of infrastructure was improved, as shown in Fig. 6 (i). In the third pair, a clear relationship between psycho-physiological restoration and overall environmental quality was not observed, the scope was identified as important in the former, as shown in Fig. 6 (j). In the second and third pairs, the explanatory power of the set of canonical variables for physiological restoration factor per se was low (9 % and 5 %, respectively), and the explanatory power of overall environment quality was very low (1 % in each of the pairs).

3.4. Effect of individual characteristics on the restoration model

An additional CCA was performed on individual characteristics and psycho-physiological responses to examine the difference in restoration response per individual differences. Table 3 shows the canonical loadings. In this round of CCA, a CCA structure comprising six models was extracted, and the participant group was segmented into six subgroups with differentiating tendencies. The six CCA models derived between individual characteristics and psycho-physiological responses showed correlations ranging from 0.41 to 0.72. However, individual characteristics explained under 2 % of the variance of psycho-physiological responses in most groups, thus showing low explanatory power.

4. Discussion

4.1. Restorative effect through urban and natural environment experience

The comparison of psycho-physiological restoration responses between urban and natural environments showed that the restoration response increased in both waterfront and green spaces compared with urban environments. This is consistent with previous findings (Deng et al., 2020; Gidlow et al., 2016; Hartig, Mang, & Evans, 1991; Park, Lee, Jung, & Swenson, 2020). As seen in the PRSS results, when exposed to a natural environment, the participants felt a great sense of relief from their responsibilities and hurdles in urban life. As shown in the soundscape and landscape perceptions, the participants experienced high audio-visual satisfaction, reported being more attracted to nature, and experienced a feeling of openness. Such positive responses to nature were also observed by an increase in valence and a decrease in arousal and negative emotions. Interestingly, in some physiological indices (SDNN, ABR), positive restoration responses were found in urban as well as natural environments, suggesting the potential of urban environments to create a restorative effect (Jeon et al., 2021; Karmanov & Hamel, 2008). Earlier studies (Stigsdotter, Corazon, Sidenius, Kristiansen, & Grahn, 2017) have shown that HRV in urban environments did not differ much from the physiological responses in green spaces and that urban environments can also have physiological restorative effects. According to Kaplan (1995), any environment has a potential restorative quality. Hence, it is feasible to improve the health of urbanites if the urban environment is properly designed and combines diverse spaces.

4.2. Restorative environmental design with audio-visual interactions

Based on the restoration model presented above, an urban design approach to improve psycho-physiological restoration responses in urbanites can be discussed from the viewpoint of audio-visual interactions. In this process, the correlations between environmental perception and audio-visual elements in the design of urban environments are also examined (see Appendix G, Table G.1).

First, urban space planning should aim to decrease detrimental soundscapes and discomforting landscapes and increase supportive or tranquil soundscapes and attractive, spatial, or natural landscapes. The most direct approach to this is controlling artificial elements (e.g., traffic, buildings, and noise) both visually and acoustically (Deng et al., 2020; Jeon & Jo, 2020). It is crucial to provide an overall calm

environment by considering important physical parameters such as grey ratio, L_{Aeq} , loudness, and L_{A5-A95} and adjusting the parameter values in appropriate ratios and to appropriate levels. Various attempts can be made, such as controlling traffic volume, changing building density in downtown areas, and reinforcing noise criteria (construction, road traffic, music). In addition, increasing green and water ratios by appropriately providing natural elements such as parks, water fountains, and man-made rivers throughout the city may also be effective. Of the natural elements, water space was the most effective, and the second most effective approach was to provide vegetation and other natural elements such as birdsong. In summary, artificial-related audio-visual elements (e.g., traffic, human, music, and building) should be reduced and nature-related audio-visual elements (e.g., birdsongs, water, wind, and vegetation) should be increased when designing urban environments. Interestingly, the prerequisite for enhanced physiological restoration is securing a tranquil soundscape and attractive landscape in Fig. 6. Moreover, controlling and regulating the physical noise level (L_{Aeq} or L_{A5-A95}) cannot be measured to satisfy such conditions and achieve this goal. That is, it is challenging to guarantee the health of urbanites with simple regulations on noise levels in the physical dimension. This notion accords with prior studies (Aletta et al., 2016; Aletta, Kang, & Axelsson, 2016), where the reduction of noise levels does not guarantee satisfaction with the sound environment. Therefore, it is also important to consider ways to create a vibrant urban environment from the soundscape perspective, using sound as a valuable resource and holding appropriate nature-based events in the urban area to induce resident activities (Jo & Jeon, 2020). For example, as suggested by Meng and Kang (2015), providing a population density of approximately 0.1–0.25 person/m² in commercial streets would ensure acoustic comfort while offering a lively urban soundscape environment.

Further, from the audio-visual interaction perspective, reducing physical factors such as L_{Aeq} , L_{A5-A95} , and Loudness revealed that the perceived affective quality of the landscape is positive from the sound perspective. Similarly, from the visual perspective, increasing natural elements such as greenery, water, and sky ratio yields a positive effect on the perceived affective quality of the soundscape. Such tendencies were consistent for audio-visual elements. Controlling the artificial sound elements and increasing the natural sound elements exerted additional positive effects on the landscape. Conversely, controlling the artificial visual elements and adding natural visual elements yield supplementary positive effects on the soundscape. Thus, various audio and visual elements that constitute the context of a city interact. Therefore, even in situations where only one visual or auditory element can be controlled during urban planning, improvements in the opposing sensory environment can be achieved through additional effects.

A previous study (Meyer-Grandbastien, Burel, Hellier, & Bergerot, 2020) has shown that biodiversity in an urban green area may increase the psychological restoration response. If traffic and other noises are unavoidable, masking them by utilizing natural sounds is also feasible. As the psychological effect of natural sounds in humans has already been demonstrated in studies, not only a masking effect but also a restorative effect can be expected with the use of natural sounds (Alvarsson, Wiens, & Nilsson, 2010; Jeon et al., 2010; Jeon et al., 2012; Ratcliffe, Gatesleben, & Sowden, 2016; Zhao et al., 2018). The presence of humans, both visually and auditorily, can negatively impact psycho-physiological restoration responses. However, it is speculated that such a finding was observed because humans were hardly present in the natural space under evaluation, in contrast to the urban environment. An excessive level of human crowd density increases loudness and therefore is highly likely to negatively impact restoration responses. As an appropriate density level can promote comfort and the activities of people sharing the space can have positive effects, future studies should focus on the restorative effect emanating from humans (Jo & Jeon, 2020; Meng & Kang, 2015).

Consequently, this study's model has practical use for urban designers, planners, and decision-makers regarding urban design or urban

environment improvements. Each stakeholder can determine design objectives based on the model. For instance, to design a fascinating and powerful city, the physical characteristics and psychological restoration model, as in Fig. 6(a), can be used. The order of Grey ratio (-0.99) > Loudness (-0.89) > L_{Aeq} (-0.85) > L_{A5-A95} (0.59) > Water ratio (0.57) exhibits the order of importance, and city stakeholders can primarily attempt to control artificial elements through policy measures. Through policies, urban building floor area ratio and building-to-land ratio can be adjusted, and height limitations can be imposed. Moreover, planting trees in urban parks or along roadsides can decrease the relative grey ratio. Similarly, reducing traffic congestion on major roads to decrease the vehicle ratio per unit area can contribute to a pleasant urban environment. The stakeholders can then propose day and night environmental noise regulations to control loudness or L_{Aeq} . Efforts can also be made to establish regulations targeting construction noise to address the temporary increase in L_{Aeq} during building or road construction events. Controlling human movement is key to decreasing L_{A5-A95} . Hence, measures must be established that disperse high-density commercial areas into low-density ones. Thus, this study's model serves as a guideline for urban planners to approach and prioritize various elements per their design objectives. This model can provide basic data for the decision-making process of urban planners, as it easily and intuitively predicts and analyzes the psycho-physiological responses of urbanites when various auditory factors that can be used in urban planning are controlled or provided. This model can also be used to develop urban improvement plans as per the described process to ensure the sustainability of cities and enhance existing urban areas into better environments. While the physiological responses may have limitations when used for design purposes or standards, they can serve as supplementary tools to anticipate the impact of urban design on the health of urbanites.

4.3. Limitations and further research

This study presents the following limitations. First, the study was conducted with participants belonging to the age group of 20 to 29 years, creating the challenge of generalizing the findings to other age groups. The study findings did not reflect sociocultural differences, since all participants were Korean and lived in a city. This study also investigated short-term responses, indicating the necessity to investigate the long-term restorative effect of continuous exposure to stimuli. Although this study used a head-mounted display (HMD) headphone-based VR technology, one of the most popular methodologies, there is a possibility of discovering new results by providing participants with a more immersive environment if augmented reality and mixed reality technologies are applied as alternatives to further increase ecological validity (Hong et al., 2021). In particular, MR technology allows participants to interact with the virtual environment rather than simply observing the virtual space. As it can include various senses other than vision and hearing, it is expected to elicit psycho-physiological responses based on responses through the five human senses in a limited laboratory environment. Moreover, to control senses other than vision and hearing, this study evaluated conditions such as heat and odor by controlling them equally. However, it is necessary to consider thermal conditions and other senses that change over time in cities (He et al., 2022). In addition, the current study only analyzed the mean brain wave values. In future studies, it is necessary to examine α and β waves in different brain regions or functional connectivity. Moreover, it is possible to uncover a new mechanism of restoration not observed here, if various physiological responses (such as blood, hormones, and brain images) are assessed besides brain waves and HRV. Despite these limitations, this study achieved the following: 1) the restorative effect of natural and urban environments was investigated from a multisensory viewpoint of audio-visual interactions; 2) both psychological and physiological restorative effects were examined using various scales; and 3) a practical restoration model was proposed, which applies to space design.

5. Conclusions

Over the past 30 years, many countries and organizations, including WHO, have made great efforts to develop healthy and sustainable cities. Though prior considerations focused on environmental pollution such as water resources and air, the importance of noise is gaining increasing significance. In this context, the findings of this study can serve as guidelines for providing healthy and sustainable urban soundscape environments for urban city users. This study examined the psycho-physiological restorative effects of exposure to soundscapes and landscapes in diverse urban and natural environments from the viewpoint of audio-visual interactions. This study proposes a new restoration model that can be useful for creating sustainable and healthy urban environments. Natural environments had a psychological restorative effect on urbanites, such as facilitating the feeling of getting away from everyday life, and a physiological restorative effect was seen in HRV and EEG. In the urban environments, a physiological restorative effect was observed in indices like SDNN and ABR, without a significant psychological restorative effect, thus confirming that urban environments can potentially improve the health of urbanites if the proper utilization of natural elements provides a pleasant audio-visual environment. This study proposed 10 psycho-physiological recovery models based on audio-visual characteristics that can be controlled, manipulated, and applied to design a sustainable and healthy city. The models have comprehensively confirmed the negative (positive) effects of artificial-related (nature-related) audio-visual elements from prior studies. From the soundscape and landscape perspective, increasing supportive and tranquil soundscapes and attractive and spatial-natural landscapes during urban planning can enhance urbanites' mental happiness and physiological health. Furthermore, the models can quantitatively analyze and easily interpret the relationship between the sound and visual elements that make up a city, including soundscape and landscape factors and psycho-physiological response, making it useful as it enables urban planners to quantitatively understand the priority of each urban design factor. Moreover, the recovery models of this study are not merely academic discoveries but are significant in that they apply to the urban design process. This study is expected to contribute significantly to reducing the practical-academic gap that has been emphasized in soundscape research. Further, the findings can be used as basic data in developing guidelines for designing healthy environments for urbanites.

Compliance with ethical standards

The study was approved by the Institutional Review Board (HYUIRB-202,010–011) of Hanyang University. Informed consent was collected from all participants.

CRediT authorship contribution statement

Jin Yong Jeon: Funding acquisition, Writing – review & editing, Project administration, Resources, Supervision. **Hyun In Jo:** Writing – original draft, Writing – review & editing, Investigation, Methodology, Conceptualization, Data curation, Formal analysis, Software, Validation, Visualization. **Kounseok Lee:** Writing – review & editing, Investigation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Audio-visual data collection method and laboratory experiment set-up with virtual reality techniques

To create an evaluation environment that could be confined to a laboratory, audio-visual data were collected at each of the sites. Data from the nine sites were collected under the same conditions: daytime (10 am–2 pm) in May 2020 when the weather was clear and warm—the average temperature of 20.6°C degrees. Visual data were recorded in high resolution (8k) using a six-channel 360 camera (Insta 360 pro, Insta 360). Auditory data were recorded by connecting a four-channel ambisonic microphone (Soundfield SPS 200, Sound field Ltd.) to a portable sound recorder (Mixpre-6, Sound devices) with an ambisonic set-up of A-formant first-order (FOA). To minimize wind noise in audio recordings, a microphone windshield was used. To calibrate the evaluation environment in the laboratory setting to the identical sound pressure level (SPL) in the field, simultaneous recordings were made using an additional microphone (half-inch microphone, AE 46, GRAS Sound and Vibration) that was connected to another portable sound recorder (AS-70, RION) (See the Fig. A.1). The camera and the microphones were all mounted on a tripod positioned in a fixed location, and all recordings were made at the average human eye level of 1.6 meters for three minutes. To synchronize the visual and auditory recordings, a clapping motion and sound were used at the start of each recording.

VR technology was used to create an evaluation environment in the laboratory, similar to the field, simulating this environment using Unity 3D software. The ecological validity of this technology in environmental evaluation has been confirmed in previous studies (Hong et al., 2019; Maffei, Masullo, Pascale, Ruggiero, & Romero, 2015).

The six-channel videos were edited to 360° images using stitching software (Insta 360 pro stitcher), and the edited visual information was presented to the participants via an HMD (VIVE Pro Eye, HTC). The four-channel sound sources were converted from A-format to B-format FOA using ambisonic plug-ins (sound devices), and the output was downmixed to binaural, by Spatial audio SDK (Spatial Audio ALI, Google) built in Unity 3D software. The participants received edited auditory information through an open-type headphone (HD-650, Sennheiser). Sound orientation was implemented in real-time, according to the changing head position of the participants turning their heads, by using a tracker built into the HMD. To adjust the sound of the VR environment to the SPL of actual locations, nine binaural sound sources were recorded using a head and torso simulator (Type 4100, Brüel & Kjær). The average SPL of the two-channel sound sources was adjusted, to match the average SPL of the sound in the field (L_{Aeq}), by using Adobe Audition software (version 1.5, Adobe). All stimuli were edited to last three minutes.



Fig. A.1. Image of hardware for (a) audio-visual data collection and (b) virtual reality environment reproduction.

Appendix B. Physical characteristics of the audio-visual stimulus

The physical characteristics of the on-site audio-visual environment were classified as acoustic or visual parameters and were quantified as shown in Table B.1. Acoustic parameters included sound strength (L_{Aeq}), spectral contents ($L_{Ceq-Aeq}$), temporal variation (L_{A5-A95}), and psychoacoustics (loudness and sharpness). L_{Aeq} indicated A-weighted equivalent continuous sound pressure levels. $L_{Ceq-Aeq}$ was defined as the difference between C-weight SPL and A-weight SPL, while L_{A5-A95} was the difference between the 5th and 95th percentile sound pressure levels. Loudness referred to the subjective perception of sound pressure strongly associated with SPL. Sharpness was determined based on the distribution of a sound's high- and low frequencies, for a specific value of loudness. The impact of a high-frequency spectrum causes rising sharpness with increasing volume. The impact analysis was performed using dBFA32 software (01 dB) and SQT Sound Quality software (PULSE, Brüel & Kjær). among visual parameters, the color ratio was analyzed for green, water, sky, and grey, representing vegetation (e.g., tree, forest, and grass), waterfront (e.g., river, ocean, water and fountain), sky, and artifacts (e.g., building, road, and vehicle), respectively, through the following steps. First, 3D videos were converted to 2D images. The researchers then used Photoshop to manually divide the outer boundary for each visual factor. As in Fig. B.1, each segmented object was assigned a layer for further differentiation and segmentation. Thus, to avoid potential errors from algorithm-based image segmentation techniques, the researchers separated the objects personally. Next, an image color summarizer (Krzywinski, 2003) computed the ratio of a segment to the entire pixel area, and these ratios were added for each of the green, water, sky, and grey elements. Luminance was also analyzed (Puyana-Romero, Ciaburro, Brambilla, Garzón, & Maffei, 2019) by computing an RGB ratio on unsegmented 2D images, as shown in Fig. B.1, using the image color summarizer. Subsequently, the standard luminance formula computed the luminance (International Telecommunications Union, 2011) - $0.299 \times \text{red} + 0.587 \times \text{green} + 0.114 \times \text{blue}$.

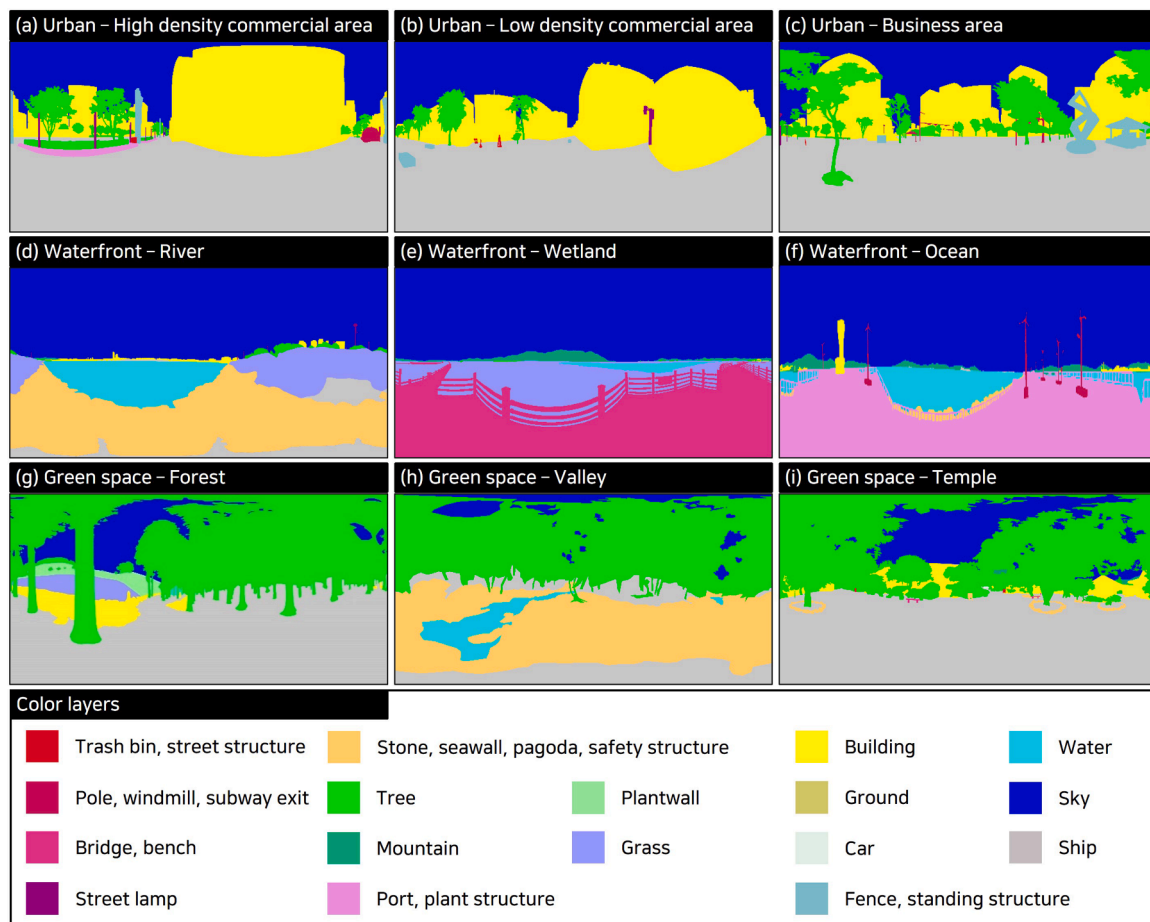


Fig. B.1. Image segmentation of nine evaluation sites: (a) – (c) urban area, (d) – (f) waterfront area, and (g) – (i) green area.

The resulting acoustic characteristics of the selected sites were as follows: L_{Aeq} ranged from 44.3–75.9 dB, $L_{Ceq-Aeq}$ ranged from 6.3–27.2 dB, and L_{A5-A95} ranged from 5.2–21.5 dB. Thus, each of the parameters showed a max-min difference of over 10 dB, indicating that the stimuli sufficiently covered variations in the typical outdoor sound setting. Loudness ranged from 5.2–78.1 sone, showing a trend similar to L_{Aeq} , while sharpness ranged from 0.9–2.1 acum. The urban environment showed a high grey ratio, the waterfront showed a high sky ratio, and the green space showed a high green ratio. Luminance ranged from 113 to 162 Cd/m^2 . The physical indices indicated that the selected sites sufficiently covered the audio-visual characteristics of various urban and natural environments.

Table B.1

Physical characteristics of the nine evaluation site.

Parameter	Acoustic parameter					Visual parameter				
	L_{Aeq} [dB]	$L_{Ceq-Aeq}$ [dB]	L_{A5-A95} [dB]	Loudness [sone]	Sharpness [acum]	Green [%]	Water [%]	Sky [%]	Grey [%]	Luminance [Cd/m^2]
Urban										
(a) High-density commercial area	75.9	12.4	16.5	39.2	1.3	7	0	18	73	141
(b) Low-density commercial area	70.6	11.5	17.4	59.6	1.4	3	0	25	70	129
(c) Business area	74.2	12.9	20.6	78.1	1.1	14	0	18	62	138
Waterfront										
(d) River	50.7	27.2	10.3	7.6	2.0	8	8	45	0	162
(e) Wetland	45.4	16.4	7.1	20.8	2.1	10	1	47	0	149
(f) Ocean	44.3	12.7	5.2	5.2	1.9	0	6	47	0	162
Green space										
(g) Forest	48.9	10.1	11.6	9.0	1.1	40	2	7	0	113
(h) Valley	67.2	6.3	10.0	20.3	1.3	41	4	4	0	142
(i) Temple	51.7	16.6	21.5	16.6	0.9	37	0	10	3	133

Appendix C. Detailed individual characteristics of participants

Regarding health state, 56 participants were classified into perceived stress scale (PSS) mid or PSS high, indicating that the participant group was at

least moderately stressed (Lee et al., 2012). That is, most participants were highly stressed with a strong need to destress and possibly required professional intervention. Regarding noise sensitivity, the total scores based on 21 items showed a wide range: 45 to 112.

Temperament was examined in four dimensions (novelty seeking, NS; harm avoidance, HA; reward dependence, RD; persistence, P), and character, three dimensions (self-directedness, SD; cooperativeness, C; self-transcendence, ST; Min et al., 2007). Temperament refers to inherited dispositions and emotional tendencies, and character refers to unique personalities or individualities acquired via environmental influences and relationships. Generally, temperament is interpreted based only on NS, HA, and RD; thus, P was excluded from the score for temperament. T scores for temperament and character were categorized into three types using three cut-off criteria (High, H: $T \geq 55$, Medium, M: $45 \leq T \leq 55$, Low, L: $T \leq 45$). Regarding temperament, the mean T scores of NS, HA, and RD were profiled as H-M-M; the participant group could be described as high novelty seeking. They were curious about new or unfamiliar stimuli and would actively explore them, were frank about their feelings, were interested in a wide array of topics, and enjoyed challenges. Regarding character, the mean T scores of SD, C, and ST were profiled as M-M-M. The group could be described as intermediate adaptive optimum. Overall, they were confident but would often give up long-term goals for practical and immediate satisfaction, sometimes showing a behavior pattern of pursuing their interests, though capable of sympathizing with others. That is, the participant group showed strong individualistic tendencies while considering rationality and objectivity as important. At an individual level, the group comprised participants with diverse temperaments and character types.

Life satisfaction was examined over four domains (physical, psychological, social, and environmental). Total scores were obtained by first calculating the mean score across four domains per the scoring method suggested by the WHOQOL Group (1998) and multiplying the outcome by 4. The range of the total scores was 42.8–86.6, with a mean of 68.9, indicating that, overall, the participant group had high life satisfaction. All mean domain scores were above the midpoint, 10, confirming that participants were relatively highly satisfied with their lives across the domains. The psychological security generally experienced by urbanites was assessed with the Urban Residents Psychological Security scale, and the mean scores of the following four domains were examined: self-psychological security (SPS), social security (SS), social environmental risk perception (SERP), and natural environmental security. The mean score was the highest in SS; thus, participants were quite satisfied with the economic development level, employment, and need satisfaction in the city they lived in. Moreover, the mean SPS and SERP scores were high, suggesting that participants were optimistic about interpersonal relationships and future life and were not too concerned about finance.

Table C.1
Summary of individual characteristics.

Parameter	Mean (SD)	Min	Max
Age (number of participants)	24.3 (2.4)	21	30
Male (30)	25.5 (2.3)	21	30
Female (30)	23.2 (1.9)	21	28
Health state (number of participants)			
Perceived Stress Scale (PSS)	21.7 (4.2)	12	30
- PSS normal: 0 – 12 (3)	12.0 (0.0)	12	12
- PSS low: 13 – 15 (1)	14.0 (0.0)	14	14
- PSS mid: 16 – 18 (7)	17.3 (0.0)	16	18
- PSS high: 19 – 40 (49)	23.0 (3.1)	19	30
Noise sensitivity	79.1 (16.1)	45	112
Temperament and character (TCI-RS) (number of participants)			
Novelty Seeking (NS) (Low: 5, Medium: 19, High: 36)	58.5 (10.0)	40.0	88.0
Harm Avoidance (HA) (Low: 20, Medium: 22, High: 18)	49.4 (12.2)	22.0	73.0
Reward Dependence (RD) (Low: 12, Medium: 20 High: 28)	54.9 (13.9)	23.0	89.0
Persistence (P)	51.4 (10.7)	31.0	75.0
Self-Directedness (SD) (Low: 12, Medium: 26, High: 22)	51.6 (11.2)	17.0	77.0
Cooperativeness (C) (Low: 19, Medium: 20, High: 21)	49.8 (13.4)	1.0	77.0
Self-Transcendence (ST) (Low: 26, Medium: 19, High: 15)	46.8 (9.2)	29.0	70.0
Quality of life (WHOQOL-BREF)			
Overall quality of life and general health	14.2 (2.7)	8.0	20.0
Domain 1: Physical domain	13.0 (1.6)	9.7	19.4
Domain 2: Psychological domain	13.4 (2.3)	7.3	19.3
Domain 3: Social domain	14.3 (2.9)	4.0	20.0
Domain 4: Environmental domain	14.0 (2.5)	9.0	19.0
Total score	68.9 (8.9)	42.8	86.6
Urban Residents Psychological Security (URPS)			
Self-psychological security (SPS)	3.2 (0.9)	1.2	5.0
Social security (SS)	3.4 (0.6)	2.2	5.0
Social environmental risk perception (SERP)	3.1 (0.8)	1.0	4.8
Natural environmental security (NES)	2.9 (1.1)	1.0	5.0

Appendix D. Detailed account of the experimental procedure

On Day 1, psycho-physiological responses were measured as follows. When the participants arrived at the laboratory, a short session was conducted to explain the previously distributed written documents to aid them in fully understanding the questionnaire items. Subsequently, the participants put on the VR devices (HMD and headphones) and the hardware for measuring physiological responses (HRV and EEG) (See the Fig. D.1). The functioning of the devices and any signal interference were checked while the participants were wearing them. For participants unfamiliar with VR devices, a brief training session (e.g., how to choose an answer to a question, how to submit an answer, and how to replay a stimulus) was provided to acclimatize them to the evaluation environment.

Once all preparations were completed, the participants rested for a short while to make themselves comfortable, and their physiological responses in a resting state were measured. These measurements in a resting state were used as a reference to compare the physiological responses to the urban, waterfront, and green space experiences. The comparison between the state of stimulus and no stimulus (an environment with a black background and

with no audio-visual information) was necessary to establish the significance of the effect of the audio-visual stimuli. Accordingly, physiological responses were measured for three minutes each in the stress task and the reference resting state, and a questionnaire was administered regarding the reference (Table 1-II) to determine the reference values of HRV, EEG, and psychological restoration responses. When physiological responses during the resting state were measured, the participants were instructed to minimize body movements and sit in a comfortable position.

When pre-test measurements were complete, stress task–audio-visual stimuli were presented for three minutes each. HRV and EEG were also measured for three minutes. After the physiological responses were measured, the same audio-visual stimuli were presented simultaneously once more, and the participants were instructed to respond to the psychological restoration questionnaire (Table 1-II). These processes made up one session. Upon completing a session, the participants took a one-minute break before the same processes were repeated for a different audio-visual stimulus.

On Day 2, environmental perception concerning soundscape and landscape was assessed using the following procedure. Unlike on Day 1, on Day 2 the participants put on the VR devices (HMD and headphones) only. The objective on Day 2 was to obtain the participants' evaluation of the audio-visual environment rather than restoration responses, and, thus, the stress task was omitted. After one of the same audio-visual stimuli as on Day 1 was presented for three minutes, the stimulus was presented again, and the participants were instructed to respond concerning environmental perception (Table 1-III). These processes constituted a session, and after a one-minute break, the same processes were repeated for a different audio-visual stimulus. The order of the stimulus environment provided on Day 2 was randomly arranged such that the participants experienced the audio-visual environment differently relative to Day 1 to eliminate bias caused by effect. The experiment was designed such that on both days, the participants were able to complete the questionnaire within the VR environment, and the data were automatically saved on the server. A stimulus was repeatedly presented if the participants requested it while answering the questionnaire. In consideration of participant fatigue and the side effects of using VR devices (e.g., cyber nausea and dizziness) that might occur during a long-running experiment, the experiment was designed to be completed within approximately one hour on Day 1 and 45 minutes on Day 2. To this end, the participants were shown six different stimuli (two each of urban, waterfront, and green space) randomly selected from a total of nine, in random order. Thus, for each stimulus the data were collected from 40 participants, and, consequently, 360 data points (nine sites $\times$ 40 participants) were obtained for each questionnaire item and each physiological indicator.

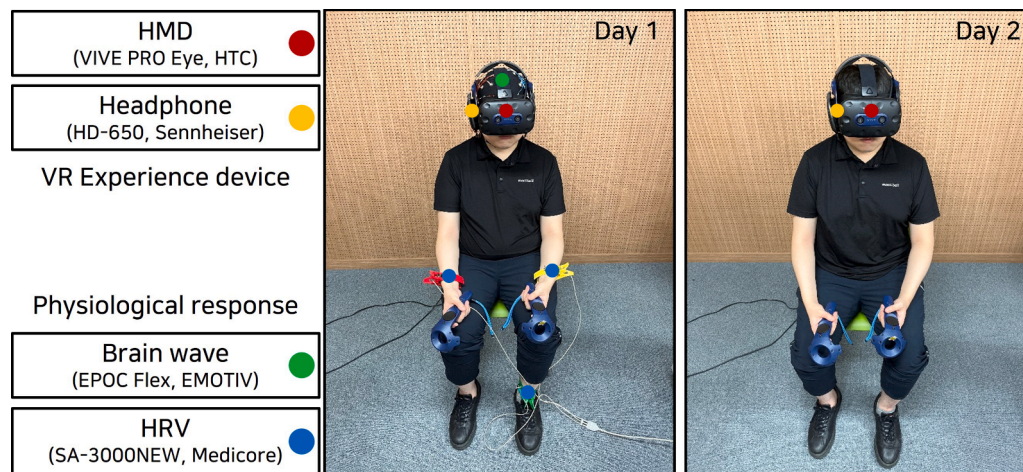


Fig. D.1. Experiment set-up with virtual reality device and physiological response measurement device.

Appendix E. Questionnaires

First, the category of individual characteristics comprised items regarding the state of their health and those regarding demographic characteristics, including a simple question regarding gender. The state of their health was assessed with 1) the perceived stress scale, a scale assessing the subjective stress level in everyday life, and 2) items regarding sensitivity to noise. Demographic characteristics were assessed with 1) the Temperament and Character Inventory (Revised Short Version), a scale to assess character and temperament, 2) the World Health Organization Quality of Life (Brief Version), a scale to assess life satisfaction, and 3) the Urban Residents Psychological Security scale, a scale to assess psychological security in an urban environment typically experienced by urbanite participants.

Next, to examine the psychological restoration response, the second category consisted of 1) the Perceived Restorativeness Soundscape Scale (PRSS), developed based on ART to assess the restorative effect of soundscape in six domains: fascination, “breaking away,” “escaping to,” compatibility, coherence, and scope, 2) the Self-Assessment Manikin (SAM), a scale to assess affective response to stimuli in three domains: valence, arousal, and dominance, and 3) the Positive Affect and Negative Affect Scale (PANAS), to assess mood state in the two dimensions of positive and negative affect. Two different affective scales were used because the PANAS is a verbal assessment tool and accordingly, individuals may have different interpretations based on emotional and cultural differences in nuance and intensity. By contrast, the SAM is a pictorial assessment tool and can thus be used to control the difference in responses due to the use of words.

Lastly, the perception of the audio-visual environment was assessed separately for soundscape and landscape. Regarding the evaluation of soundscapes, within the large framework of the ISO 12,913–2 (Hotelling, 1935) protocols, the following items were used: 1) sound source identification (traffic noise, other noise, human sound, music, birdsong, water sound, wind sound), 2) 14 semantic expressions (pleasant, vibrant, eventful, informative, annoying, stressful, chaotic, artificial, uninformative, calm, relaxing, natural, monotonous, uneventful) to assess the perceived affective quality of soundscape, and 3) overall impression, appropriateness, and sound satisfaction concerning overall environmental quality. Landscapes were evaluated using the same method as for soundscapes, as follows: 1) visual elements (vehicle, building, people, open space, vegetation, water, sky), 2) 14 semantic attributes (appealing, harmonious, orderly, repulsive, disharmonious, disorderly, complex, simple, open, natural, wide, enclosed, artificial, narrow) to assess the perceived affective quality of the landscape, and 3) scenic beauty, the status of infrastructures, and visual satisfaction

concerning overall environmental quality. The number of items to evaluate soundscape and landscape, respectively, was the same to minimize interpretive errors in statistical analysis that may occur because of a different number of independent variables.

All survey items utilized in this study, except for PANAS (reliability for Koreans: 0.83, Lee, Kim, & Lee, 2003), were not validated specifically with Koreans. However, they were selected and used because their utilization as evaluation tools was deemed to have been verified by prior studies. Moreover, PRSS and PANAS were found to be highly reliable with Cronbach's alpha of 0.73 ~ 0.93 and 0.81 ~ 0.89, respectively, for each evaluation domain.

Appendix F. Principal component analysis of perceived affective quality

A principal component analysis (PCA) was performed on the responses to the 14 semantic expressions to extract the dimensions of the soundscape and landscape perception before interpreting the perceived affective quality. The PCA results are summarized in Table F.1. All Kaiser-Meyer-Olkin measures of sampling adequacy were over 0.8, suggesting that the sample was adequate and the results of Bartlett's test of sphericity were significant ($p < 0.01$). Three dimensions were extracted for soundscape perception: *supportive soundscape* (16.27 %), *detrimental soundscape* (29.27 %), and *tranquil soundscape* (28.15 %). The *supportive soundscape* was labeled Factor 1, based on the high factor loadings of attributes like vibrant and pleasant, and the *detrimental soundscape* was labeled Factor 2, based on the high factor loadings of negative attributes such as annoying and stressful. The *tranquil soundscape* was labeled as Factor 3, based on high factor loadings for monotonous and calm. Three dimensions of landscape perception were extracted: *attractive landscape* (19.25 %), *discomforting landscape* (21.44 %), and *spatial/natural landscape* (21.83 %). The *Attractive landscape* was labeled Factor 1, given the high factor loadings of orderly, appealing, and harmonious, and the *discomforting landscape* as Factor 2, given the high factor loadings of disharmonious and repulsive. Similarly, the *spatial and natural landscape* was labeled Factor 3, with high factor loadings of simple, open, natural, and wide.

The factor scores were computed using the extracted soundscape and landscape dimensions. Factor score is a numerical value of a variable that belongs to a particular factor, and in this study, the scores were computed using regression analysis (Var, 1998).

Table F.1

Principal component analysis for semantic expressions of soundscape and landscape.

Attributes Soundscape ^{a)}	Factor 1 Supportive (16.27)	Factor 2 Detrimental (29.27)	Factor 3 Tranquil (28.15)	Cronbach Alpha
Pleasant	0.79	-0.16	0.19	0.67
Vibrant	0.83	0.14	-0.27	
Eventful	0.73	0.23	-0.39	
Informative	0.47	-0.28	0.47	0.76
Annoying	-0.01	0.82	-0.31	
Stressful	-0.01	0.82	-0.34	
Chaotic	0.12	0.81	-0.42	0.90
Artificial	0.09	0.65	-0.54	
Uninformative	-0.04	0.77	0.04	
Calm	-0.07	-0.51	0.73	0.90
Relaxing	0.13	-0.54	0.71	
Natural	0.05	-0.54	0.72	
Monotonous	-0.11	-0.25	0.80	0.90
Uneventful	-0.40	0.04	0.77	
Landscape ^{b)}	Attractive (19.25)	Discomforting (21.44)	Spatial/Natural (21.83)	Cronbach Alpha
Appealing	0.63	-0.36	0.34	0.76
Harmonious	0.69	-0.33	0.25	
Orderly	0.82	-0.15	0.03	
Repulsive	-0.19	0.79	-0.28	0.83
Disharmonious	-0.20	0.81	-0.18	
Disorderly	-0.42	0.67	-0.11	
Complex	-0.24	0.57	-0.56	0.89
Simple	0.52	-0.11	0.61	
Open	0.33	-0.13	0.78	
Natural	0.50	-0.38	0.60	0.89
Wide	0.35	-0.02	0.78	
Enclosed	0.12	0.36	-0.67	
Artificial	-0.28	0.49	-0.60	0.89
Narrow	-0.01	0.39	-0.73	

^{a)} Kaiser-Meyer-Olkin (KMO) = 0.892, Chi-square: $\chi^2(91) = 3915.57$, $p < 0.01$

^{b)} Kaiser-Meyer-Olkin (KMO) = 0.871, Chi-square: $\chi^2(91) = 3481.19$, $p < 0.01$

Note: The rotation method is varimax rotation, and the eigenvalue is more than 1.

Appendix G. Pearson correlation analysis between urban design element and environmental perception

Table G.1

Pearson correlation analysis based on environmental perception.

Variables	SS	DS	TS	AL	DL	SNL	OI	AP	SOSA	SB	SI	VISA
Physical characteristics												
Acoustic - LAeq	-0.61**	0.09	0.47**	-0.33**	0.31**	-0.58**	-0.50**	-0.25**	-0.38**	-0.57**	0.41**	-0.53**
Acoustic - L _{Ceq} -Aeq	0.26**	-0.09	-0.07	0.09	-0.07	0.25**	0.16**	0.14**	0.13*	0.13*	0.10	0.15**
Acoustic - LA ₅₋₉₅	-0.41**	-0.01	0.35**	-0.19**	0.20**	-0.54**	-0.35**	-0.14**	-0.26**	-0.43**	0.33**	-0.42**

(continued on next page)

Table G.1 (continued)

Variables	SS	DS	TS	AL	DL	SNL	OI	AP	SOSA	SB	SI	VISA
Physical characteristics												
Acoustic - Loudness	-0.61**	0.04	0.52**	-0.38**	0.35**	-0.55**	-0.53**	-0.28**	-0.44**	-0.62**	0.41**	-0.58**
Acoustic - Sharpness	0.28**	-0.06	-0.15**	0.03	-0.07	0.46**	0.18**	0.09	0.10*	0.19**	-0.08	0.22**
Visual - Green ratio	0.23**	0.08	0.31**	0.27**	-0.25**	-0.01	0.29**	0.15**	0.26**	0.31**	-0.32**	0.26**
Visual - Water ratio	0.40**	0.02	0.32**	0.26**	-0.18**	0.47**	0.36**	0.19**	0.30**	0.44**	-0.17**	0.43**
Visual - Sky ratio	0.21**	-0.11*	0.04	-0.02	0.01	0.40**	0.08	0.04	0.03	0.09	0.01	0.12*
Visual - Grey ratio	-0.66**	0.04	-0.58**	-0.42**	0.39**	-0.58**	-0.59**	-0.30**	-0.46**	-0.67**	0.50**	-0.62**
Visual - Luminance	0.23**	-0.11*	0.08	0.04	-0.03	0.39**	0.10	0.04	0.06	0.14*	0.03	0.16**
Audio-visual element												
Acoustic - Traffic noise	-0.61**	0.05	-0.62**	-0.35**	0.44**	-0.54**	-0.57**	-0.31**	-0.46**	-0.65**	0.53**	-0.58**
Acoustic - Other noise	-0.52**	0.07	-0.65**	-0.27**	0.49**	-0.50**	-0.57**	-0.28**	-0.43**	-0.62**	0.47**	-0.54**
Acoustic - Human sound	-0.55**	0.14**	-0.50**	-0.24**	0.32**	-0.61**	-0.47**	-0.18**	-0.33**	-0.53**	0.39**	-0.48**
Acoustic - Music	-0.34**	0.10	-0.52**	-0.22**	0.37**	-0.36**	-0.39**	-0.22**	-0.19**	-0.46**	0.31**	-0.38**
Acoustic - Birdsongs	0.35**	0.21**	0.38**	0.37**	-0.28**	0.16**	0.43**	0.24**	0.39**	0.45**	-0.30**	0.41**
Acoustic - Water sound	0.39**	0.15**	0.36**	0.31**	-0.16**	0.45**	0.40**	0.23**	0.36**	0.52**	-0.26**	0.47**
Acoustics - Wind sound	0.55**	0.09	0.35**	0.43**	-0.23**	0.36**	0.41**	0.24**	0.37**	0.47**	-0.40**	0.45**
Visual - Vehicle	-0.58**	0.03	-0.58**	-0.32**	0.43**	-0.51**	-0.51**	-0.31**	-0.44**	-0.62**	0.52**	-0.54**
Visual - Building	-0.57**	0.03	-0.59**	-0.32**	0.42**	-0.56**	-0.54**	-0.25**	-0.40**	-0.64**	0.53**	-0.56**
Visual - People	-0.61**	0.08	-0.55**	-0.31**	0.38**	-0.63**	-0.52**	-0.24**	-0.40**	-0.63**	0.42**	-0.54**
Visual - Open space	0.40**	0.08	0.20**	0.26**	-0.04	0.54**	0.33**	0.17**	0.28**	0.37**	-0.25**	0.39**
Visual - Vegetation	0.37**	-0.01	0.24**	0.30**	-0.20**	0.18**	0.35**	0.19**	0.39**	0.34**	-0.31**	0.34**
Visual - Water space	0.49**	0.01	0.35**	0.28**	-0.18**	0.51**	0.39**	0.18**	0.30**	0.46**	-0.37**	0.44**
Visual - Sky	0.38**	0.06	0.24**	0.27**	-0.09	0.50**	0.33**	0.24**	0.30**	0.34**	-0.14**	0.38**

Note: SS: Supportive soundscape; DS: Detrimental soundscape; TS: Tranquil soundscape; AL: Attractive landscape; DL: Discomfort landscape; SNL: Spatial/Natural landscape; OI: Overall impression; AP: Appropriateness; SOSA: Sound satisfaction; SB: Scenic beauty; SI: Status of infrastructure; VISA: Visual satisfaction. * and ** represent $p < 0.05$ and $p < 0.01$, respectively.

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